Transform Yourself

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PYTHON PROGRAMMING

Python Basics

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Fundamentals of Python

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Introduction to python programming

- High level, interpreted language
- Object-oriented
- General purpose
  - Web development (like: Django, and Bottle),
  - Scientific and Mathematical Computing (Orange, SciPy, NumPy)
  - Desktop graphical user Interfaces (Pygame, Panda3D).

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Other features of python includes the following:

- Simple language which is easier to learn
- Free and open source
- Portability
- Extensible and embeddable
- Standard large library

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- Created by Guido van Rossum in 1991
- Why the name Python?
  - Not after a dangerous snake.
  - Rossum was fan of a famous TV Show comedy series from late Eighties.
  - The name "Python" was adopted from the same series "Monty Python's Flying Circus".

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Python's Benevolent Dictator For Life

"Python is an experiment in how much freedom programmers need. Too much freedom and nobody can read another's code; too little and expressiveness is endangered."

- Guido van Rossum



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Reasons to Choose Python as First Language

- Simple Elegant Syntax
- Not overly strict
- Expressiveness of the language
- Great Community and Support

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Installation of Python in Windows

- Go to Download Python page on the official site (python.org/downloads/) and click Download Python **3.9.4.**
- When the download is completed, double-click the file and follow the instructions to install it.
- When Python is installed, a program called **IDLE** is also installed along with it. It provides graphical user interface to work with Python.

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Want to help test development versions of Python? Prereleases,
Docker images
Looking for Python 2.7? See below for specific releases

- Open IDLE, Write the following code below and press enter.

```
print("Hello, World!")
```

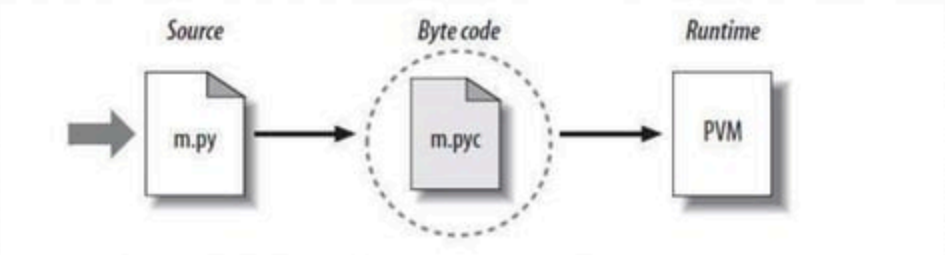
- To create a file in IDLE, go to File > New Window (Shortcut: Ctrl+N).
- Write Python code and save (Shortcut: Ctrl+S) with .py file extension like: hello.py or your-first-program.py

```
print("Hello, World!")
```

- Go to Run > Run module (Shortcut: F5) and you can see the output.

Python Code Execution

- Python's traditional runtime execution model: source code you type is translated to byte code, which is then run by the Python Virtual Machine. Your code is automatically compiled, but then it is interpreted.



Source code extension is **.py**
Byte code extension is **.pyc** (compiled python code)

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Difference Between Python 2.x & 3.x

KEY DIFFERENCE

- Python 3 syntax is simpler and easily understandable whereas Python 2 syntax is comparatively difficult to understand.
- Python 3 default storing of strings is Unicode whereas Python 2 stores need to define Unicode string value with "u."
- Python 3 value of variables never changes whereas in Python 2 value of the global variable will be changed while using it inside for-loop.
- Python 3 exceptions should be enclosed in parenthesis while Python 2 exceptions should be enclosed in notations.
- Python 3 rules of ordering comparisons are simplified whereas Python 2 rules of ordering comparison are complex.
- Python 3 offers Range() function to perform iterations whereas, In Python 2, the xrange() is used for iterations.

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Difference Between Python 2.x & 3.x



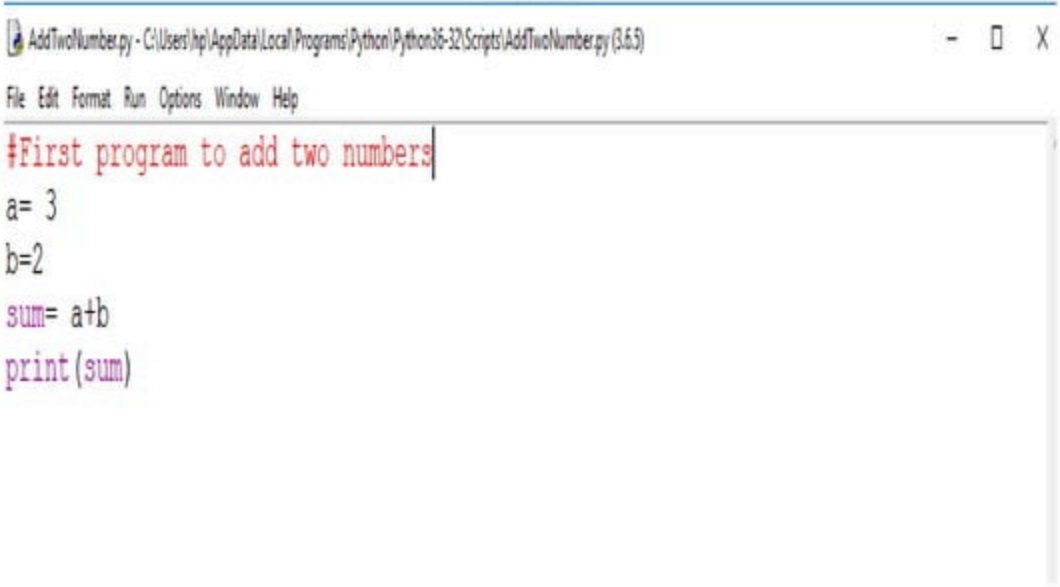
Basis of comparison	Python 3	Python 2
Release Date	2008	2000
Function print	print ("hello")	print "hello"
Division of Integers	Whenever two integers are divided, you get a float value	When two integers are divided, you always provide integer value.
Unicode	In Python 3, default storing of strings is Unicode.	To store Unicode string value, you require to define them with "u".
Syntax	The syntax is simpler and easily understandable.	The syntax of Python 2 was comparatively difficult to understand.
Rules of ordering Comparisons	In this version, Rules of ordering comparisons have been simplified.	Rules of ordering comparison are very complex.

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Difference Between Python 2.x & 3.x

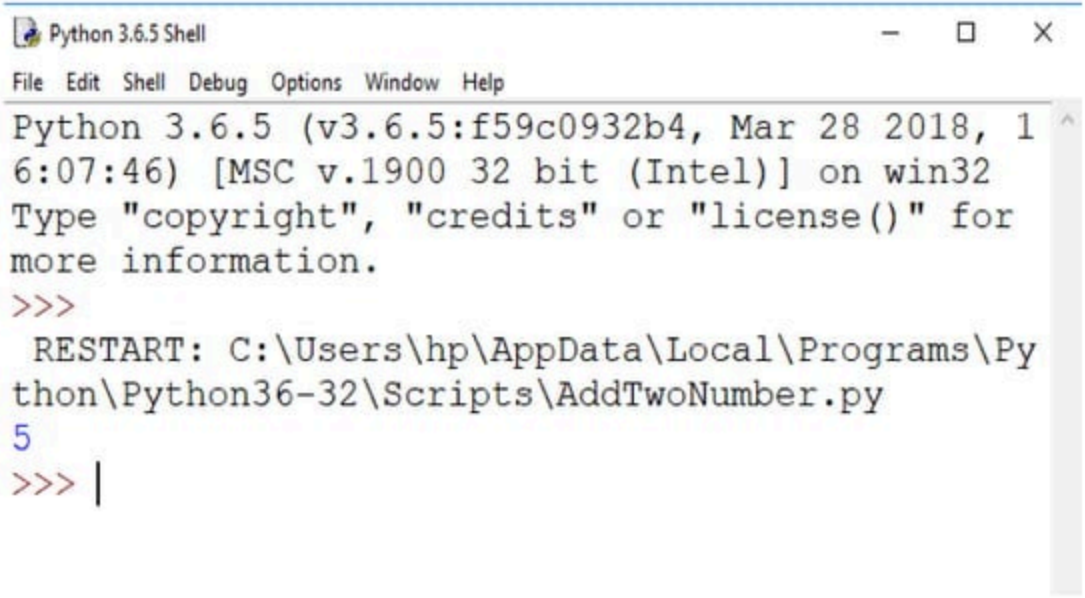
Iteration	The new Range() function introduced to perform iterations.	In Python 2, the xrange() is used for iterations.
Exceptions	It should be enclosed in parenthesis.	It should be enclosed in notations.
Leak of variables	The value of variables never changes.	The value of the global variable will change while using it inside for-loop.
Backward compatibility	Not difficult to port python 2 to python 3 but it is never reliable.	Python version 3 is not backwardly compatible with Python 2.
Library	Many recent developers are creating libraries which you can only use with Python 3.	Many older libraries created for Python 2 is not forward-compatible.

First Python Program



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Second Program

AllArithmeticOperation.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/AllArithmeticOperation.py (3.6.5)

File Edit Format Run Options Window Help

```
#Program to illustrate all the arithmetic operations

a = 21
b = 10
c = 0

c = a + b
print ("Line 1 - Value of c is ", c)

c = a - b
print ("Line 2 - Value of c is ", c )

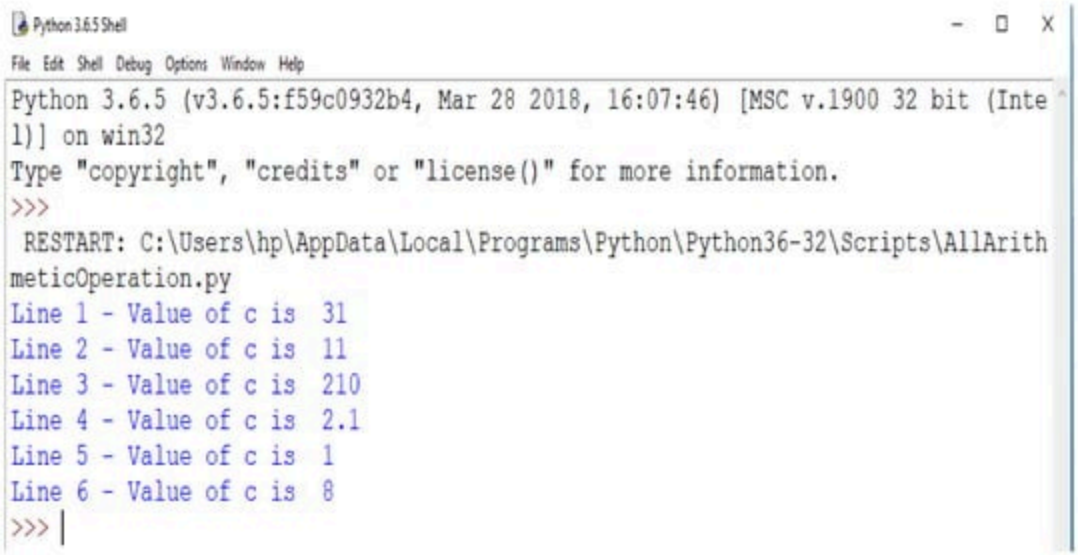
c = a * b
print ("Line 3 - Value of c is ", c )

c = a / b
print ("Line 4 - Value of c is ", c )

c = a % b
print ("Line 5 - Value of c is ", c)

a = 2
b = 3
c = a**b
print ("Line 6 - Value of c is ", c)
```

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Few Important Things to Remember

- To represent a statement in Python, newline (enter) is used.
- Use of semicolon at the end of the statement is optional (unlike languages like C/C++)
- In fact, it's recommended to omit semicolon at the end of the statement in Python
- Instead of curly braces { }, indentations are used to represent a block

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Python Keywords

- Keywords are the reserved words(can not be used as a identifier) in Python
- Are case sensitive

Keywords in Python programming language

False	class	finally	is	return
None	continue	for	lambda	try
True	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
assert	else	import	pass	
break	except	in	raise	

Python Identifiers

- Name given to entities like class, functions, variables etc. in Python
- Rules for writing identifiers
  - Can be a combination of letters in lowercase (a to z) or uppercase (A to Z) or digits (0 to 9) or an underscore (_)
  - myClass, var_1 and print_this_to_screen, all are valid examples
  - Cannot start with a digit
  - 1variable is invalid, but variable1 is perfectly fine
  - Keywords cannot be used as identifiers
  - Cannot use special symbols like !, @, #, $, % etc. in our identifier
  - Can be of any length.

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Things to care about

- Python is a case-sensitive language.
  - Variable and variable are not the same.
- Multiple words can be separated using an underscore, this_is_a_long_variable

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Python Statement

- Instructions that a Python interpreter can execute are called statements.
- Two types
  - Single line statement:
    - For example, a = 1 is an assignment statement
  - Multiline statement:
    - In Python, end of a statement is marked by a newline character.
    - So, how to make multiline statement?

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- Technique1:
  - make a statement extend over multiple lines with the line continuation character (\).
  - For example:

```
a=1+2+3+\
4+5+6+\
7+8+9
```

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- Technique2:
  - make a statement extend over multiple lines with the parentheses ( )
  - For example:

```
a=(1 +2+3+
4+5+6+
7+8+9)
```

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- Technique3:
  - make a statement extend over multiple lines with the brackets [ ] and braces { }.
  - For example:

```
colors = ['red',
          'blue',
          'green']
```

- We could also put multiple statements in a single line using semicolons, as follows

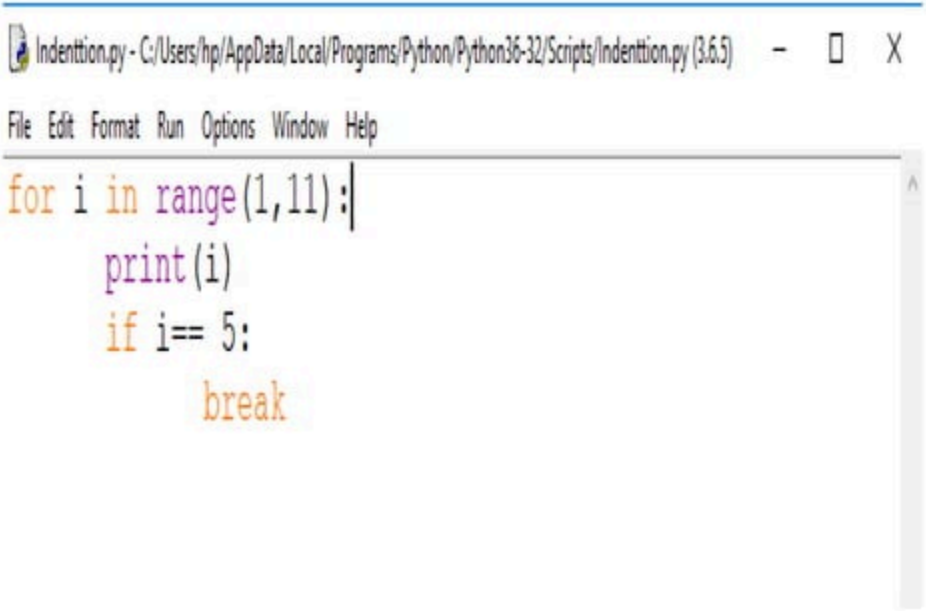
```
a=1;b=2;c=3
```

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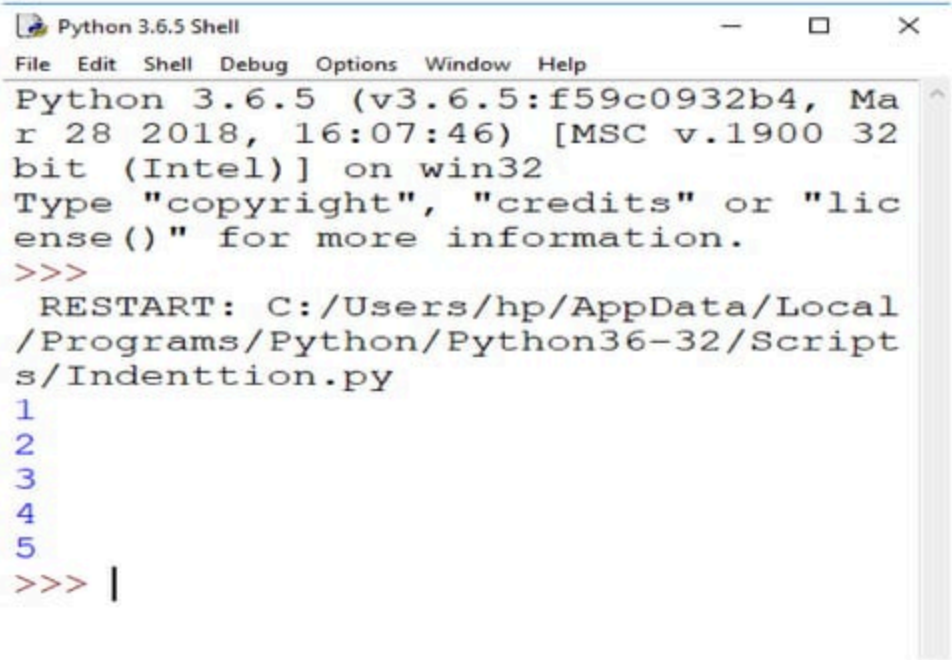
Python Indentation

- Most of the programming languages like C, C++, Java use braces { } to define a block of code
- Python uses indentation
- A code block (body of a function, loop etc.) starts with indentation and ends with the first un-indented line

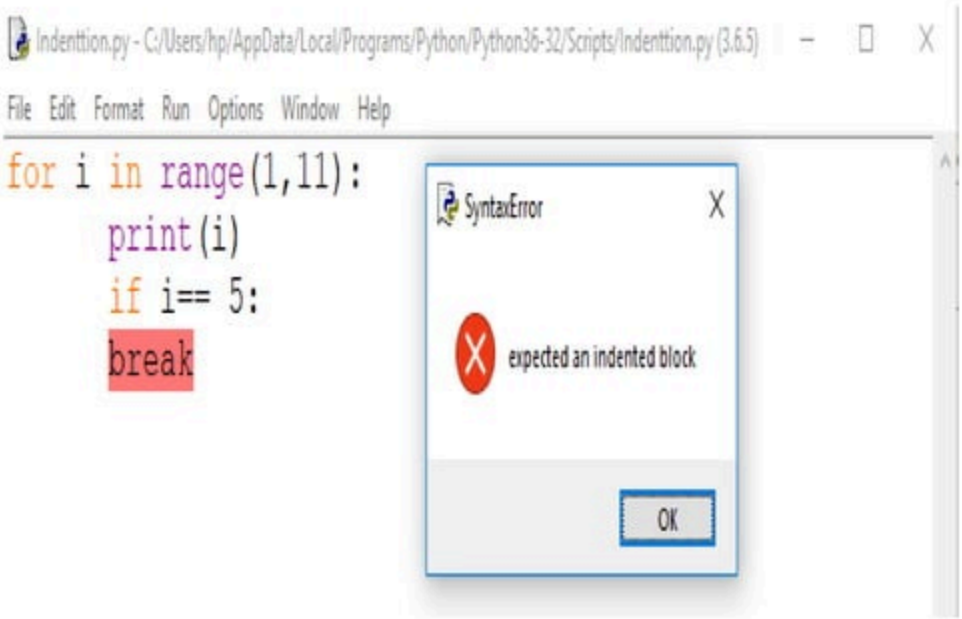
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Indenttion.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/Indenttion.py (3.6.5)
File Edit Format Run Options Window Help
for i in range(1,11):
print(i)
if i== 5:
break

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Error due to incorrect indentation



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Python Comments

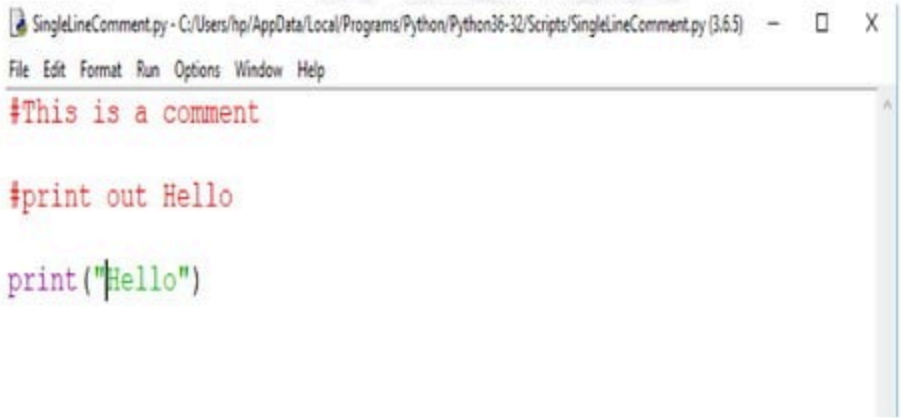
- To make the code much more readable.
- Python Interpreter ignores comment.
- Two types of comment is possible in python:
  - Single line comment and
  - Multi-line comment

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Single line comment

- In Python, we use the hash (#) symbol to start writing a comment.
- It extends up to the newline character



Multi-line comments

- Two way:
  - By using # sign at the beginning of each line
  - By using either ‘ ‘ ‘ or “ “ “ (most common)

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Python Variable

- a variable is a named location used to store data in the memory.
- Alternatively, variables are container that hold data which can be changed later throughout programming

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Declaring Variables

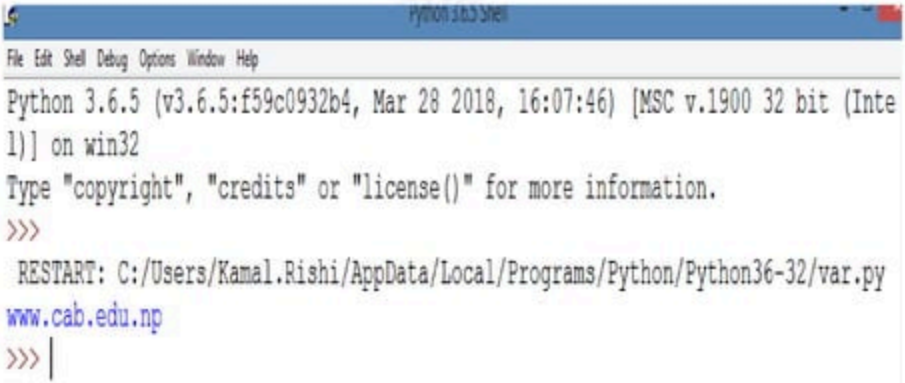
- In Python, variables do not need declaration to reserve memory space.
- The "variable declaration" or "variable initialization" happens automatically when we assign a value to a variable.
- We use the assignment operator = to assign the value to a variable.

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Assigning Values to variables




Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
RESTART: C:/Users/Kamal.Rishi/AppData/Local/Programs/Python/Python36-32/var.py
www.cab.edu.np
>>>

Changing value of a variable



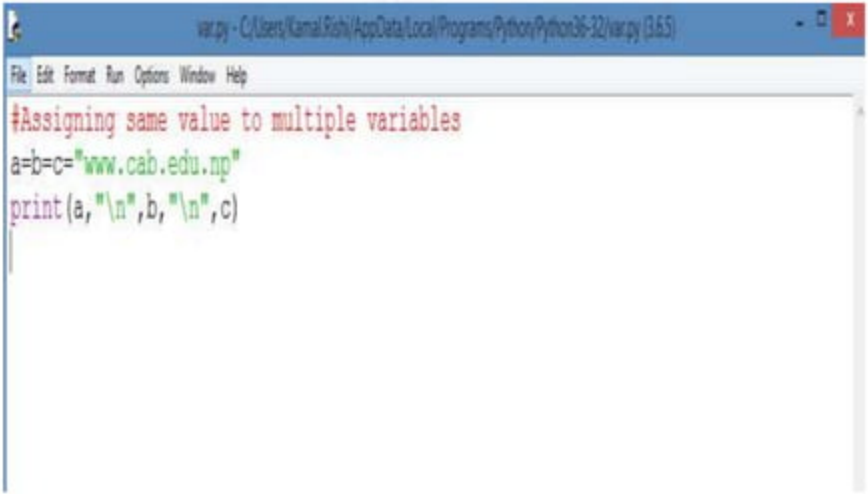
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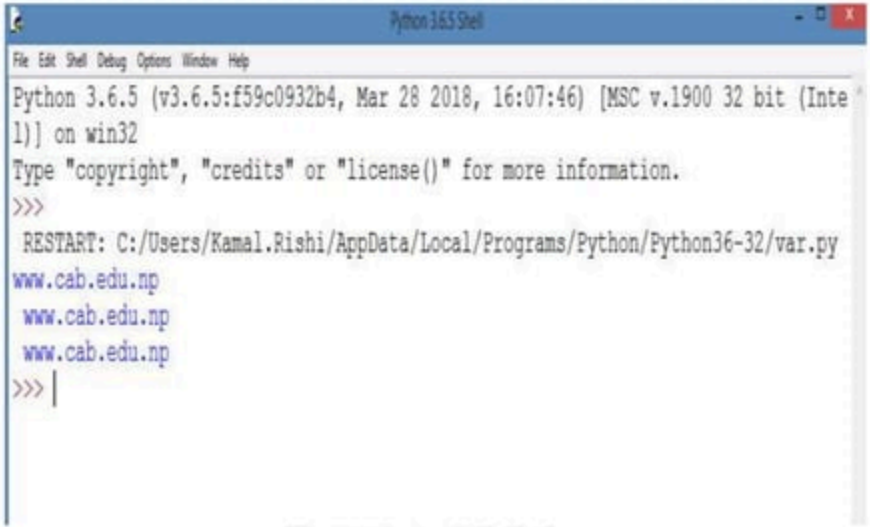


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Assigning same value to the multiple variable



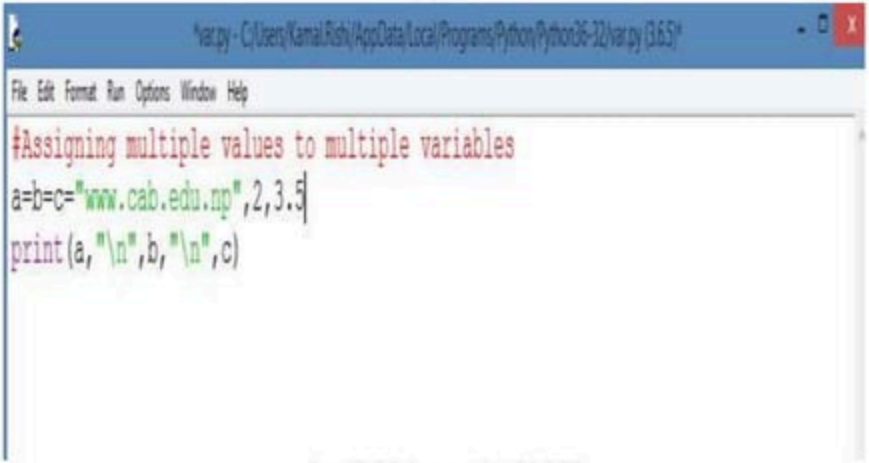
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Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46) [MSC v.1900 32 bit (Inte
l)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
RESTART: C:/Users/Kamal.Rishi/AppData/Local/Programs/Python/Python36-32/var.py
www.cab.edu.np
www.cab.edu.np
www.cab.edu.np
>>>

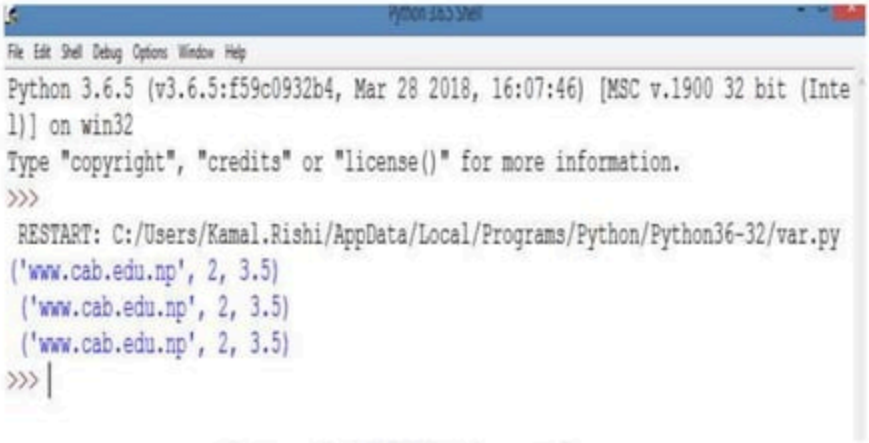
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Assigning multiple values to multiple variables



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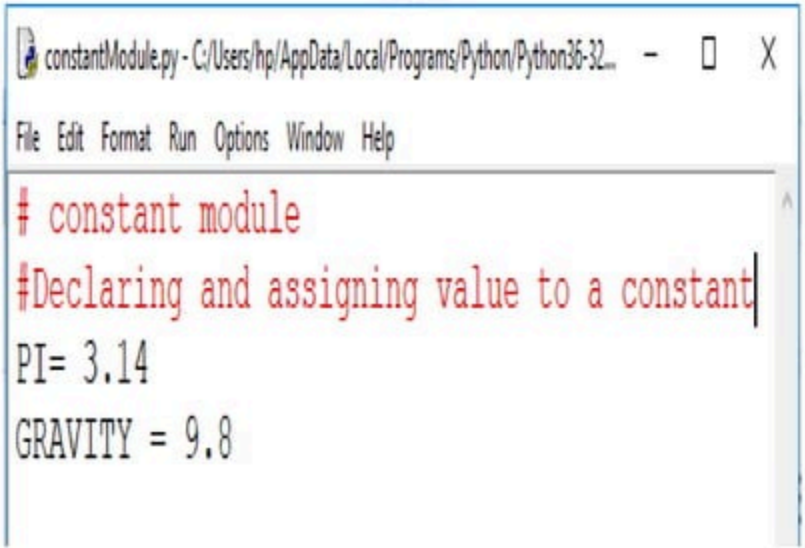
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Constants

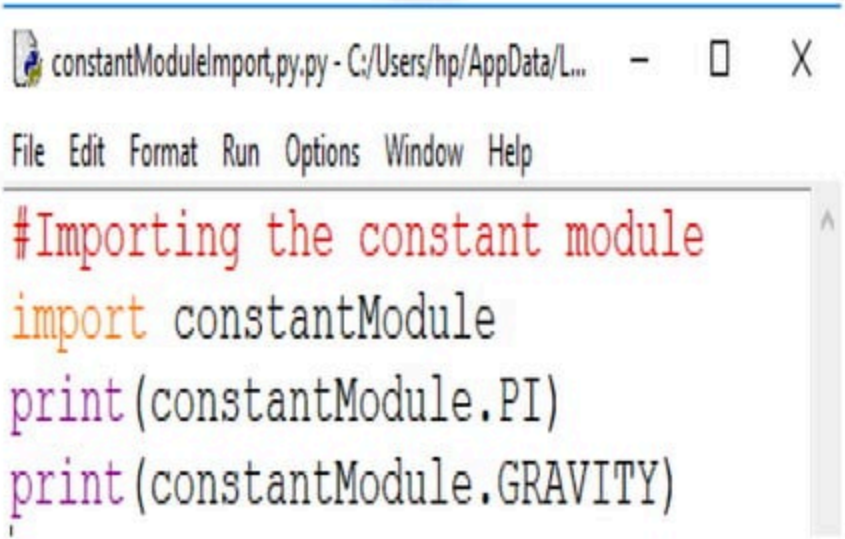
- Value that cannot be altered by the program during normal execution.
- In Python, constants are usually declared and assigned on a module
- And then imported to the file

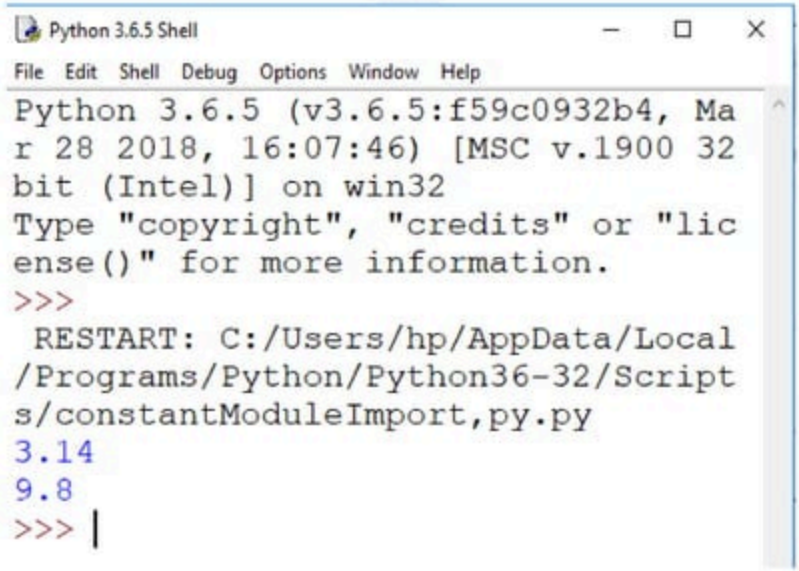
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Defining constant Module



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constantModuleImport.py.py - C:/Users/hp/AppData/L...
File Edit Format Run Options Window Help
#Importing the constant module
import constantModule
print(constantModule.PI)
print(constantModule.GRAVITY)


Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/constantModuleImport,py.py
3.14
9.8
>>>

Rules and Naming convention for variables and constants

- Create a name that makes sense. Suppose, vowel makes more sense than v.
- Use camelCase notation to declare a variable
- For example: myName, myAge, myAddress
- Use capital letters where possible to declare a constant
- For example: PI,G,MASS etc.

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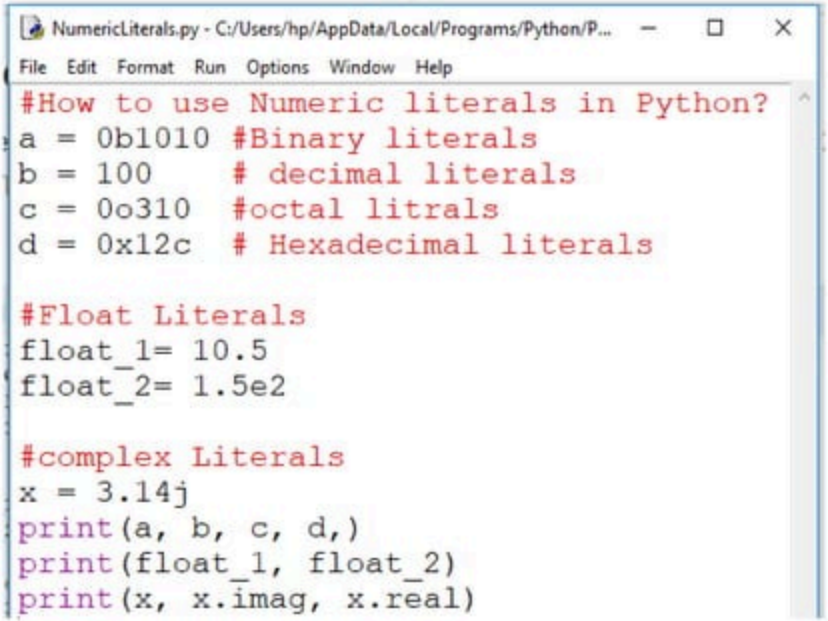
- Never use special symbols like !, @, #, $, %, etc.
- Don't start name with a digit.
- Constants are put into Python modules
- Constant and variable names should have combination of letters in lowercase (a to z) or uppercase (A to Z) or digits (0 to 9) or an underscore (_).
- For example: snake_case, MACRO_CASE, camelCase, CapWords

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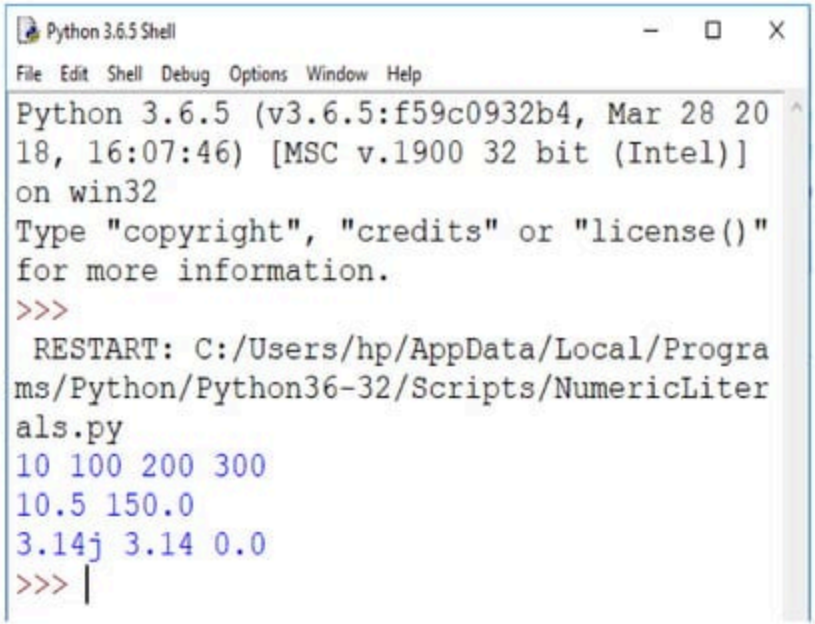
Literals

- Literal is a raw data given in a variable or constant.
- In Python, there are various types of literals they are as follows:
  1. Numeric literals – Interger, Float, Complex
  2. String literals – String, Character
  3. Boolean literals – True or False
  4. Special literals – None(Null Value)
  5. Literal collections – List, Tuple, Dictionary and Set

Numeric Literals



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Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 20
18, 16:07:46) [MSC v.1900 32 bit (Intel)]
on win32
Type "copyright", "credits" or "license()"
for more information.
>>>
RESTART: C:/Users/hp/AppData/Local/Progra
ms/Python/Python36-32/Scripts/NumericLiter
als.py
10 100 200 300
10.5 150.0
3.14j 3.14 0.0
>>>

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String Literals

StringLiteral.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/StringLiteral.py (3.6.5)

File Edit Format Run Options Window Help

```
#How to use string literals in Python?
#String
strings = " THis is python"
print(strings)
strings = 'I am new in python'
print(strings)
strings = """ But I have now problem"""
print(strings)

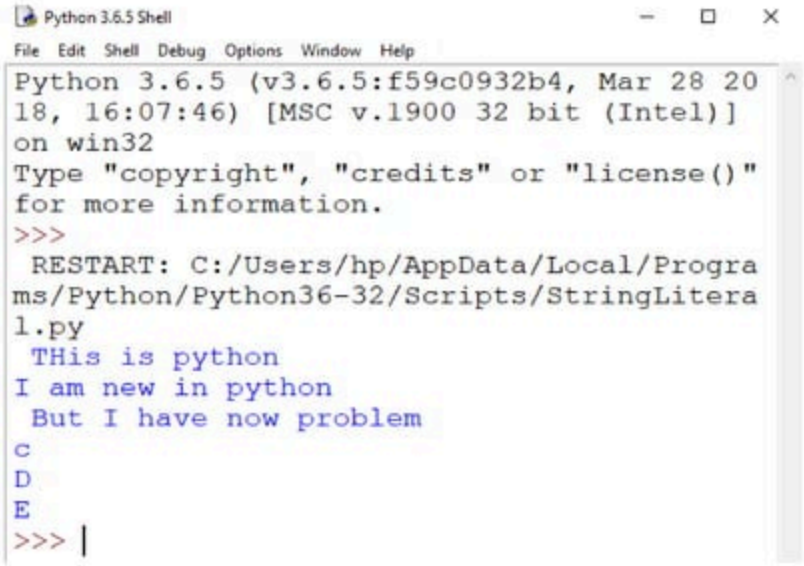
#character

char = 'c'
print(char)

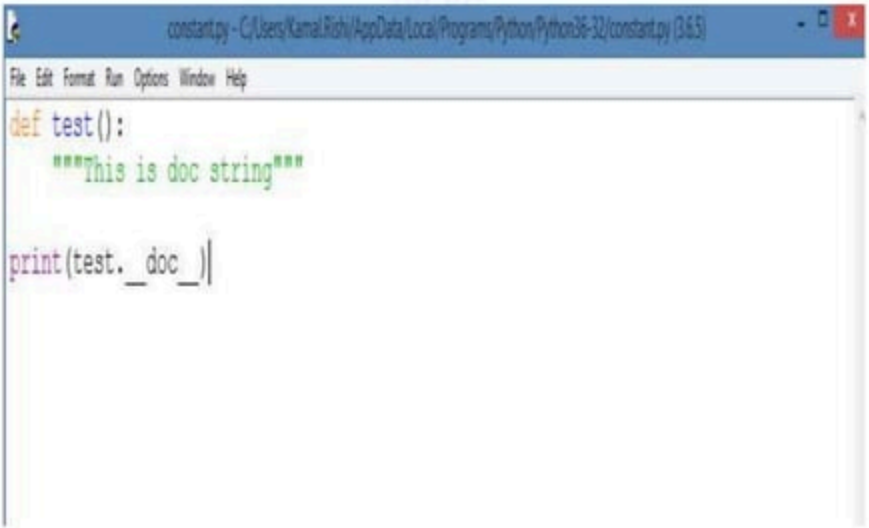
char = "D"
print(char)

char = "E"
print(char)
```

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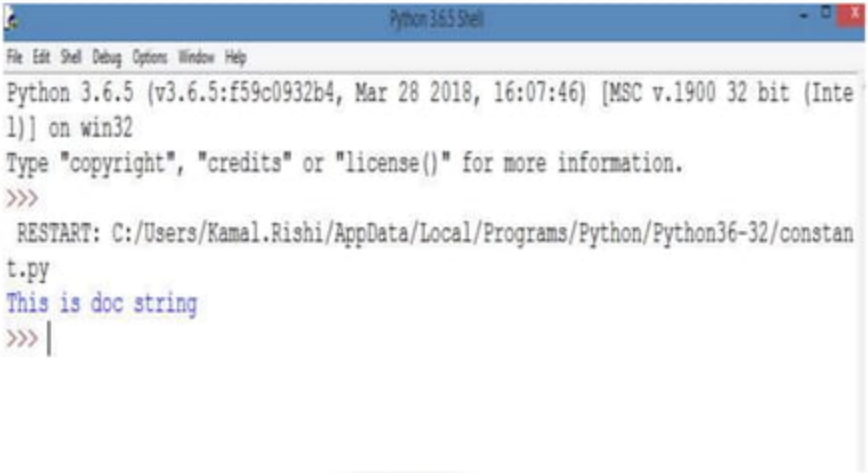


Docstrings



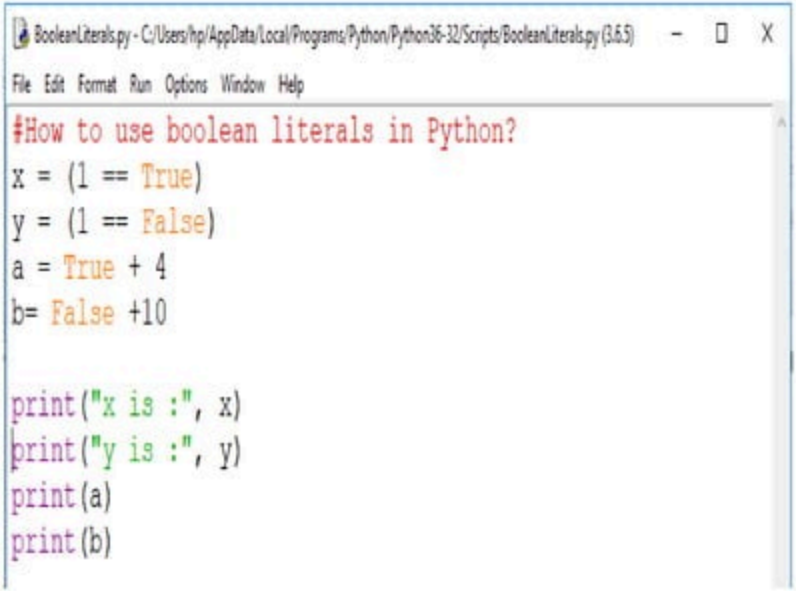
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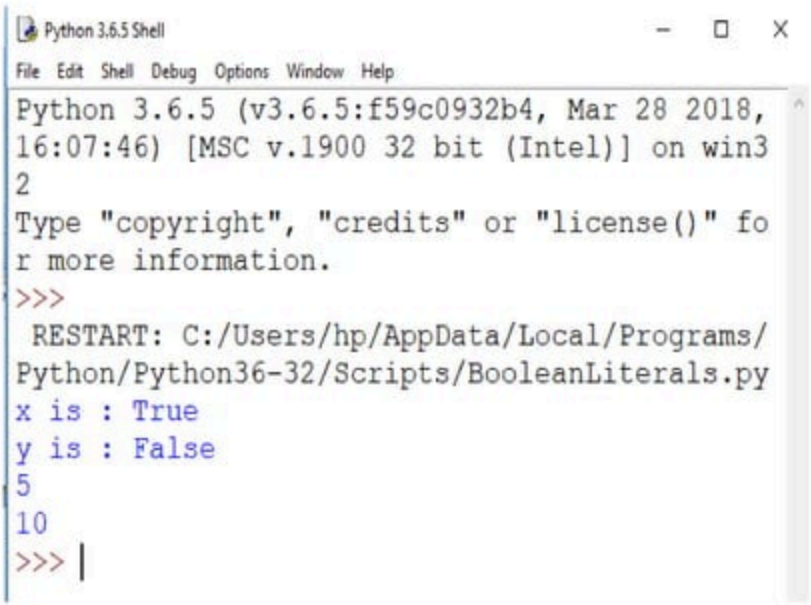
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Using boolean literals

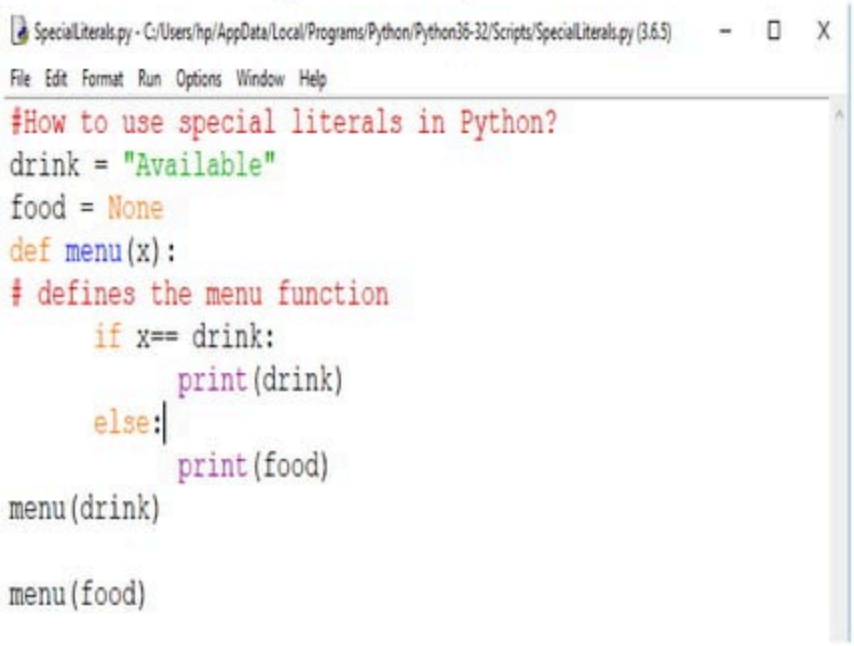


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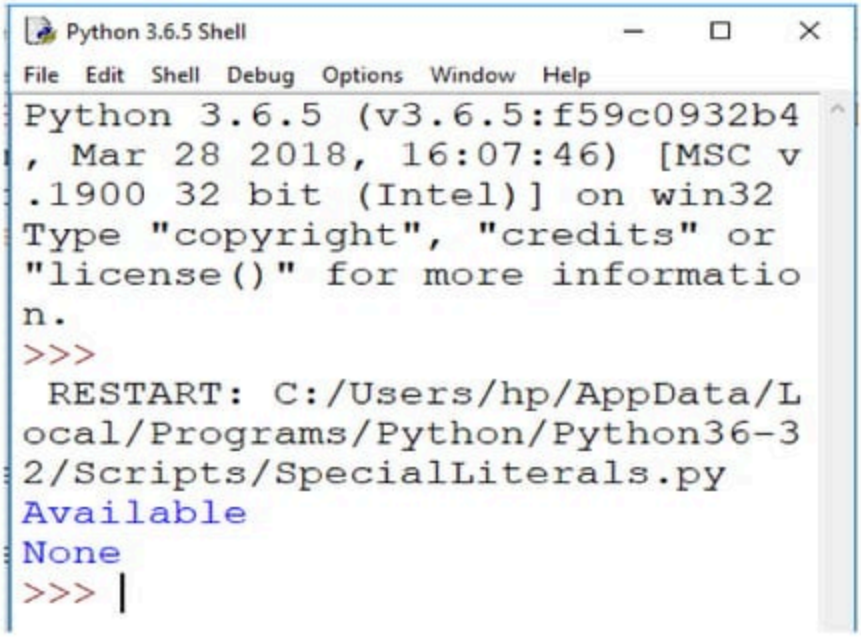
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Special literal(None)

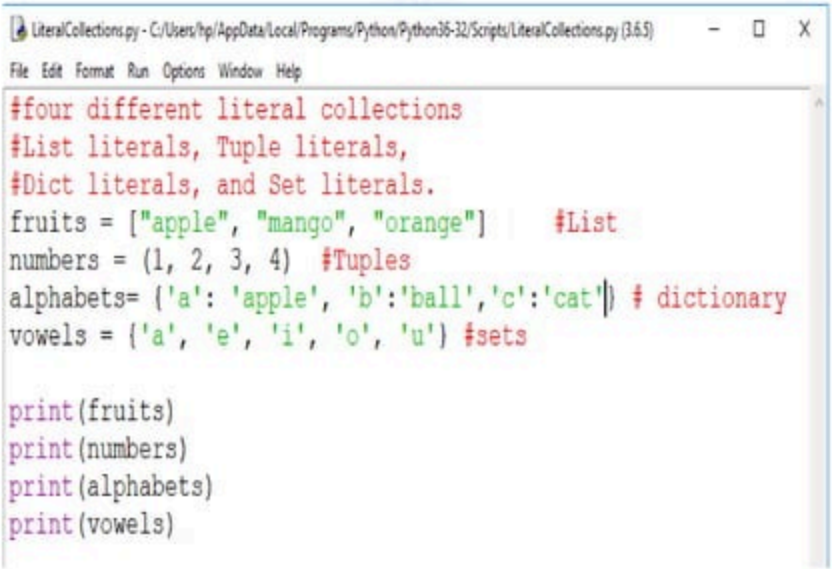


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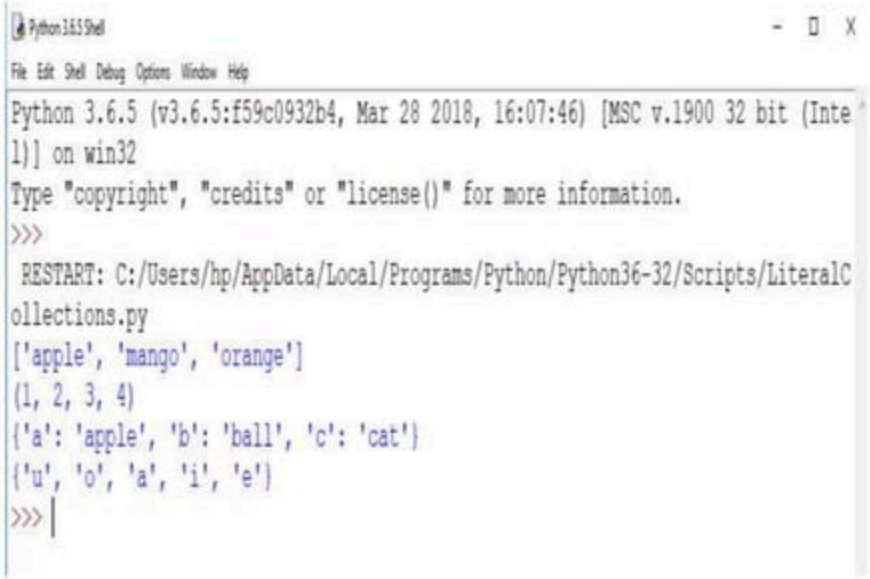

Python 3.6.5 Shell
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, Mar 28 2018, 16:07:46) [MSC v
.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or
"license()" for more informatio
n.
>>>
RESTART: C:/Users/hp/AppData/L
ocal/Programs/Python/Python36-3
2/Scripts/SpecialLiterals.py
Available
None
>>>

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Literals Collection



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Built-in Data Types

- In programming, data type is an important concept.
- Variables can store data of different types, and different types can do different things.
- Python has the following data types built-in by default, in these categories:

Text Type:	str
Numeric Types:	int, float, complex
Sequence Types:	list, tuple, range
Mapping Type:	dict
Set Types:	set, frozenset
Boolean Type:	bool
Binary Types:	bytes, bytearray, memoryview

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Setting the Data Type

➢In Python, the data type is set when you assign a value to a variable:

Example	Data Type
`x = "Hello World"`	str
`x = 20`	int
`x = 20.5`	float
`x = 1j`	complex
`x = ["apple", "banana", "cherry"]`	list
`x = ("apple", "banana", "cherry")`	tuple
`x = range(6)`	range
`x = {"name" : "John", "age" : 36}`	dict
`x = {"apple", "banana", "cherry"}`	set
`x = frozenset({"apple", "banana", "cherry"})`	frozenset
`x = True`	bool
`x = b"Hello"`	bytes
`x = bytearray(5)`	bytearray
`x = memoryview(bytes(5))`	memoryview

Setting the Specific Data Type

➢If you want to specify the data type, you can use the following constructor functions:

Example	Data Type
`x = str("Hello World")`	str
`x = int(20)`	int
`x = float(20.5)`	float
`x = complex(1j)`	complex
`x = list(("apple", "banana", "cherry"))`	list
`x = tuple(("apple", "banana", "cherry"))`	tuple
`x = range(6)`	range
`x = dict(name="John", age=36)`	dict
`x = set(("apple", "banana", "cherry"))`	set
`x = frozenset(("apple", "banana", "cherry"))`	frozenset
`x = bool(5)`	bool
`x = bytes(5)`	bytes
`x = bytearray(5)`	bytearray
`x = memoryview(bytes(5))`	memoryview

Data Types

- In python data types are classes and variables are instance (object) of these classes
- Different types of data types in python are:
  - Python numbers
  - Python lists
  - Python tuples
  - Python strings
  - Python sets
  - Python dictionary

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Python Numbers

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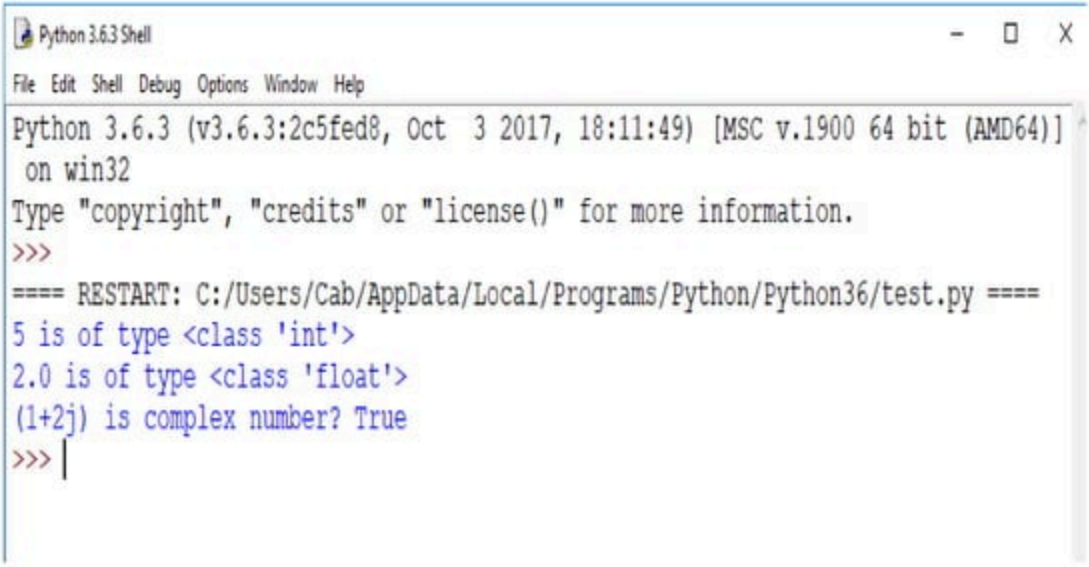
test.py - C:/Users/Cab/AppData/Local/Programs/Python/Python36/test.py (3.6.3)

File Edit Format Run Options Window Help

```
a = 5
print(a, "is of type", type(a))

a = 2.0
print(a, "is of type", type(a))

a = 1+2j
print(a, "is complex number?", isinstance(a,complex))
```


Python 3.6.3 Shell
File Edit Shell Debug Options Window Help
Python 3.6.3 (v3.6.3:2c5fed8, Oct 3 2017, 18:11:49) [MSC v.1900 64 bit (AMD64)]
on win32
Type "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:/Users/Cab/AppData/Local/Programs/Python/Python36/test.py ====
5 is of type <class 'int'>
2.0 is of type <class 'float'>
(1+2j) is complex number? True
>>>

Python List

- Ordered sequence of items
- Can contain heterogeneous data
- Syntax:
  - Items separated by commas are enclosed within brackets [ ]
  - Example: a= [1,2.2,'python']

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Extracting elements from the list

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test.py - C:/Users/Cab/AppData/Local/Programs/Python/Python36/test.py (3.6.3)

File Edit Format Run Options Window Help

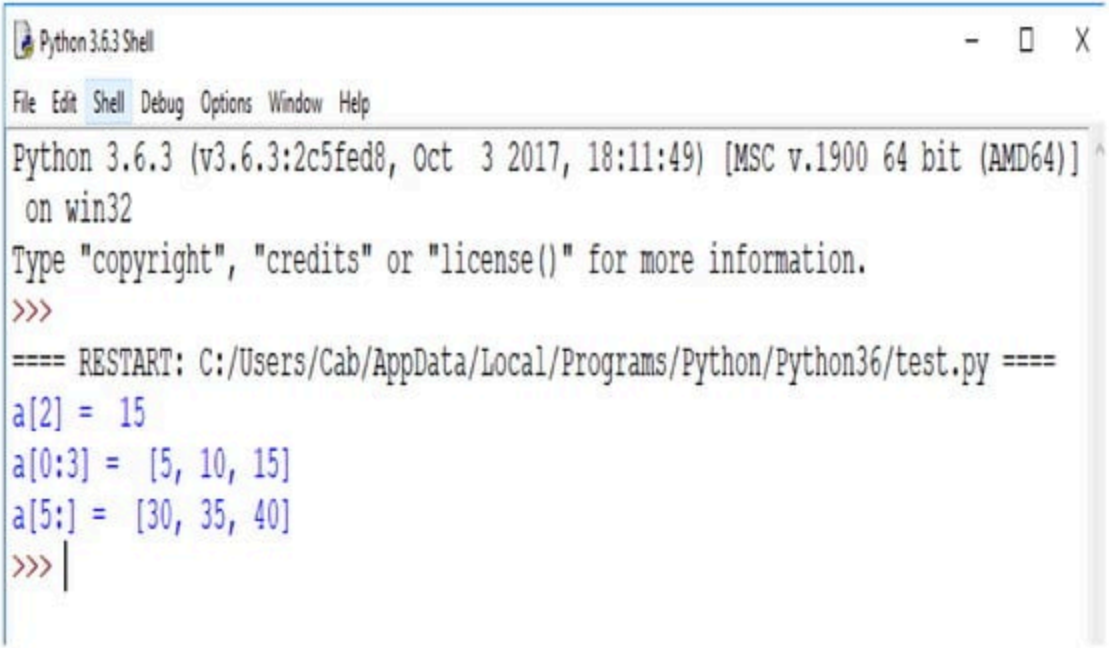
```
a = [5,10,15,20,25,30,35,40]

print("a[2] = ", a[2])

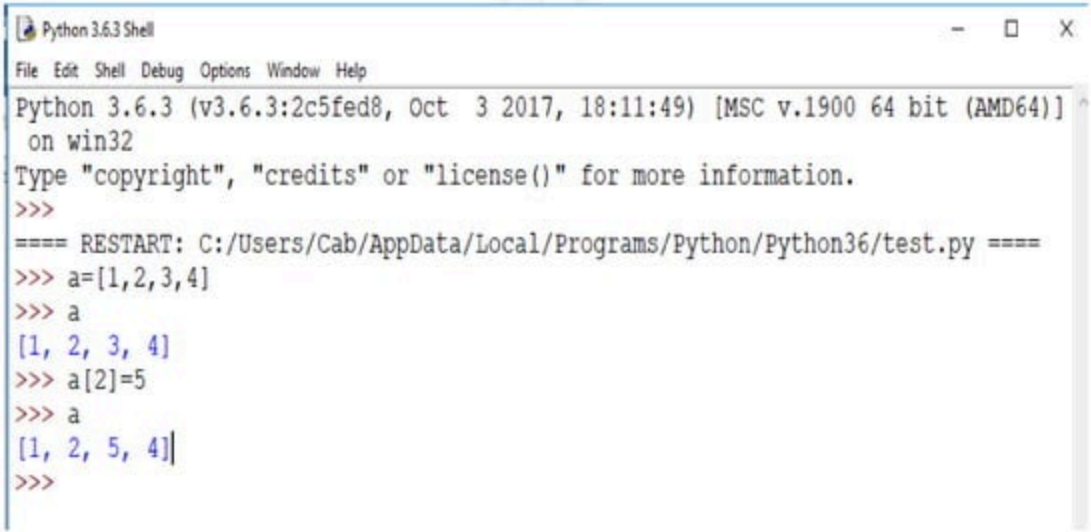
print("a[0:3] = ", a[0:3])

print("a[5:] = ", a[5:])
```

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- Note: Lists are mutable, meaning, value of elements of a list can be altered.



Python Tuple

- Same as list but immutable.
- Used to write-protect the data
- Syntax:
  - Items separated by commas are enclosed within brackets ( )
- Example:
  - t = (5,'program', 1+3j)

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*ExtratingTupleItem.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-...

File Edit Format Run Options Window Help

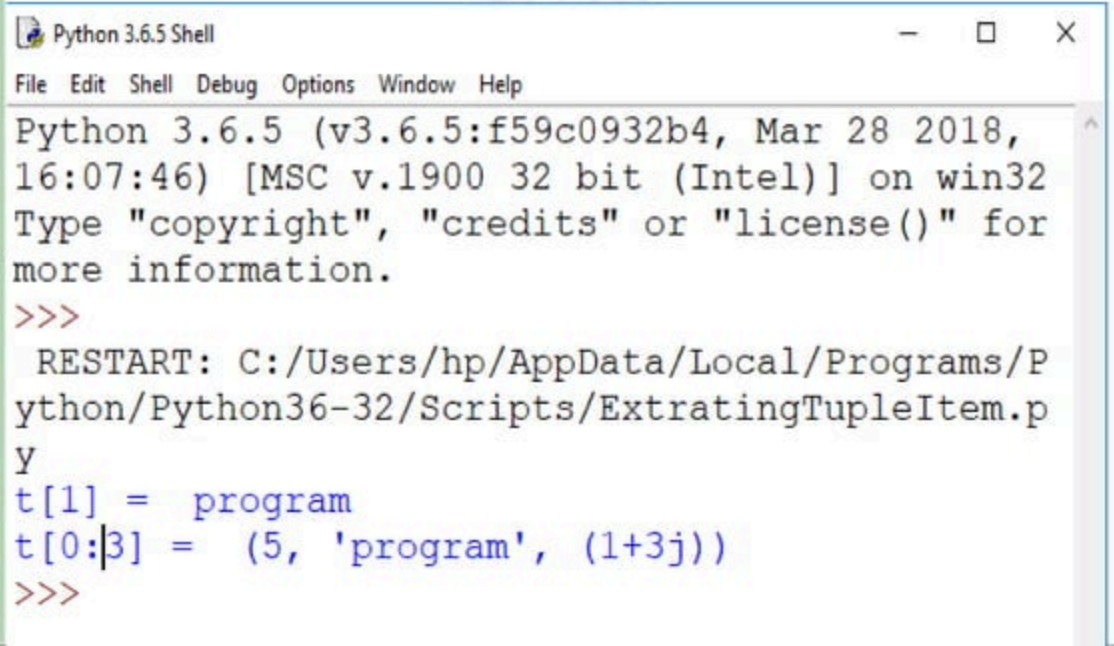
```
#Extracting the elements of the Tuple and


t = (5,'program', 1+3j)

# t[1] = 'program'
print("t[1] = ", t[1])

# t[0:3] = (5, 'program', (1+3j))
print("t[0:3] = ", t[0:3])
```

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ImmutabilityOfTuple.py - C:/Users/hp/AppData/Local/Programs/Python/Python36...

File Edit Format Run Options Window Help

```
#Extracting the elements of the Tuple and
# immutability of tuple

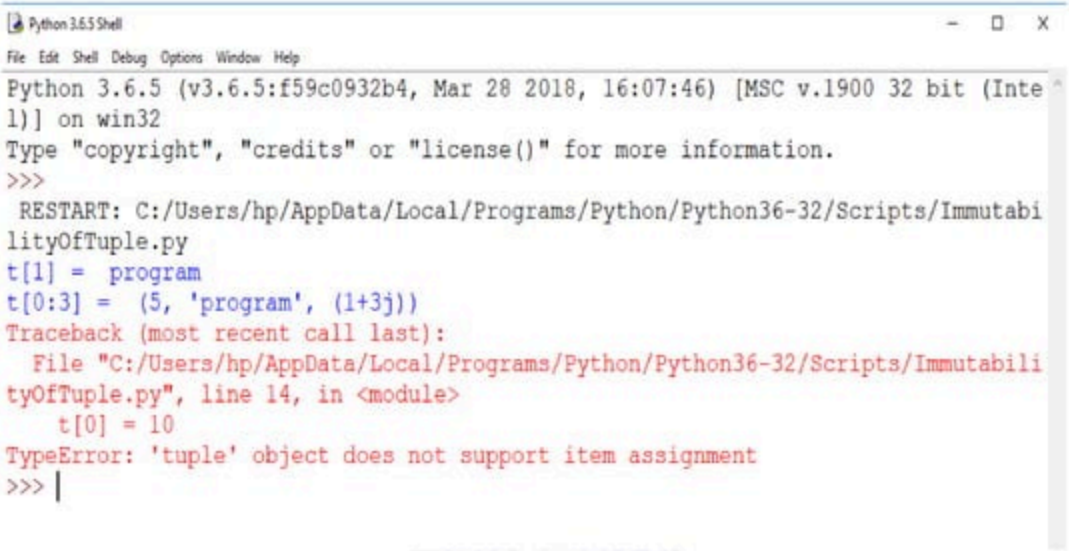
t = (5,'program', 1+3j)

# t[1] = 'program'
print("t[1] = ", t[1])

# t[0:3] = (5, 'program', (1+3j))
print("t[0:3] = ", t[0:3])

# Generates error
# Tuples are immutable
t[0] = 10
```

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Python Strings

- use single quotes or double quotes to represent strings.
- Multi-line strings can be denoted using triple quotes, ''' or """.

Example:

```
s = "This is a
string"  s = '''a
multiline
        string'''
```

- Strings are also immutable.

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PythonString.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/PythonString.py (...

File Edit Format Run Options Window Help

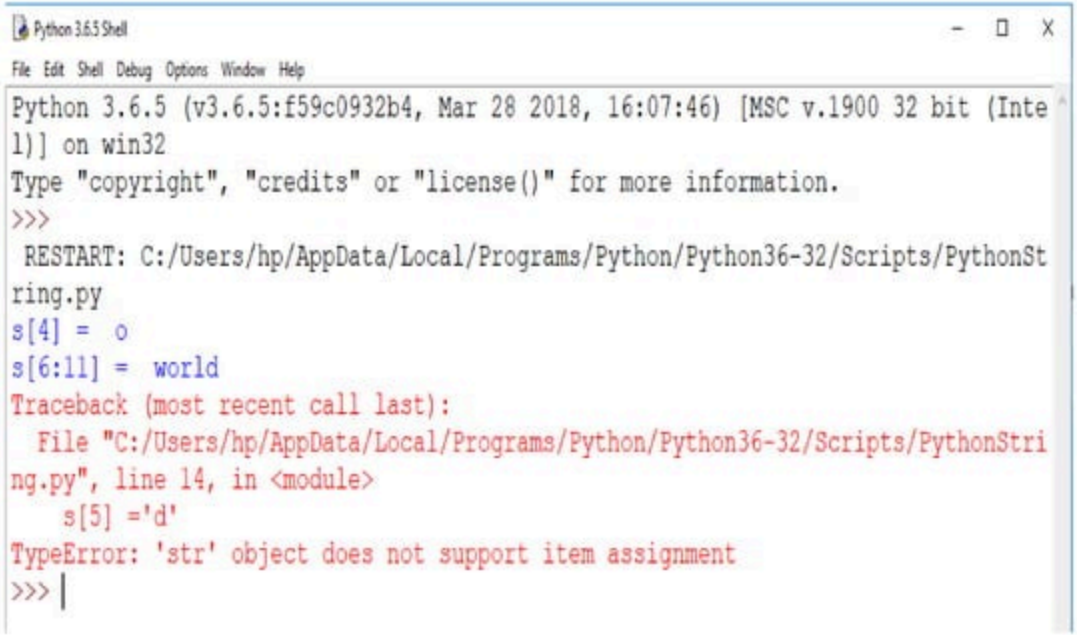
```
#python string data extraction and
#immutability of the python strings

s = 'Hello world!'

# s[4] = 'o'
print("s[4] = ", s[4])

# s[6:11] = 'world'
print("s[6:11] = ", s[6:11])

# Generates error
# Strings are immutable in Python
s[5] ='d'
```


Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46) [MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>>
RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/PythonString.py
s[4] = o
s[6:11] = world
Traceback (most recent call last):
File "C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/PythonString.py", line 14, in <module>
s[5] ='d'
TypeError: 'str' object does not support item assignment
>>>

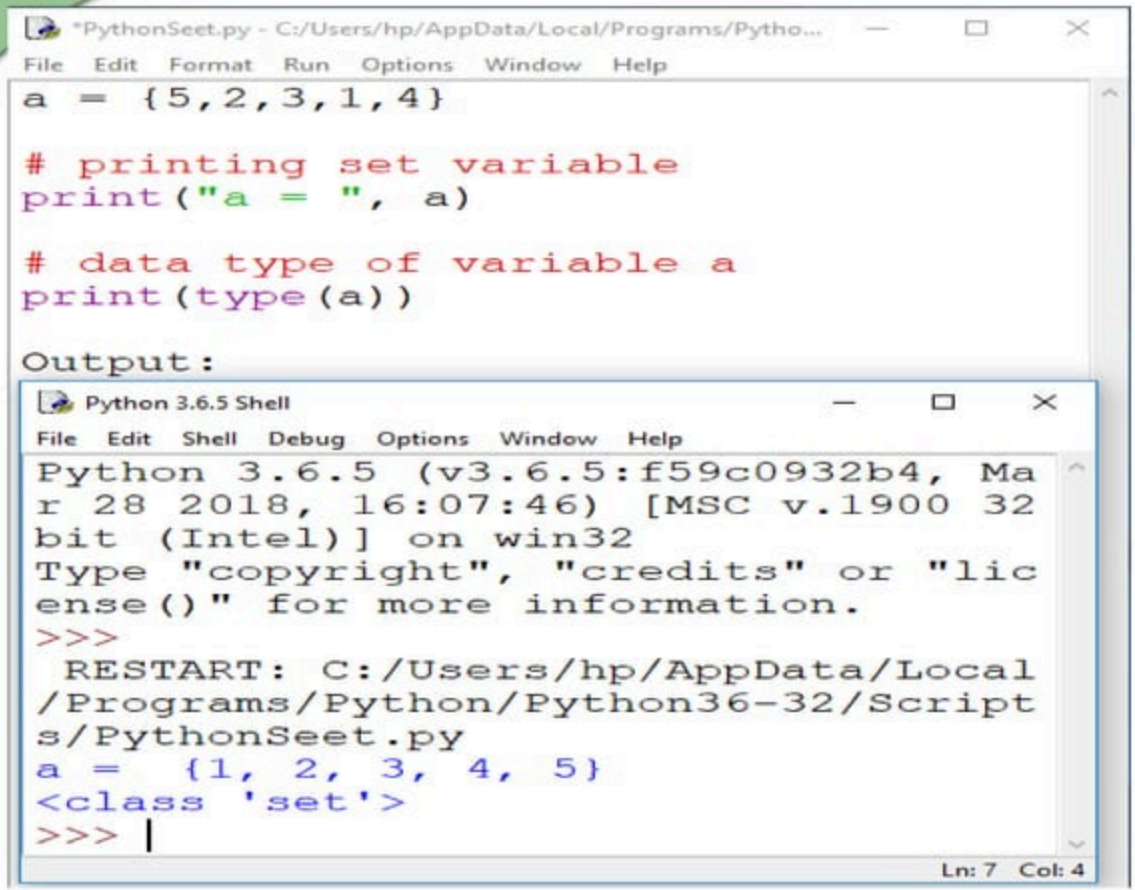
Python Set

- Unordered collection of unique items
- defined by values separated by comma inside braces { }

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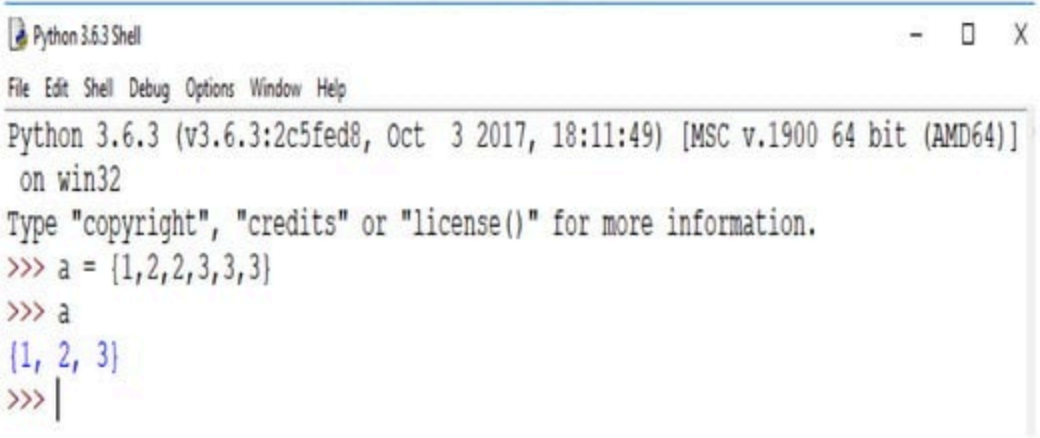
Estd : 1984

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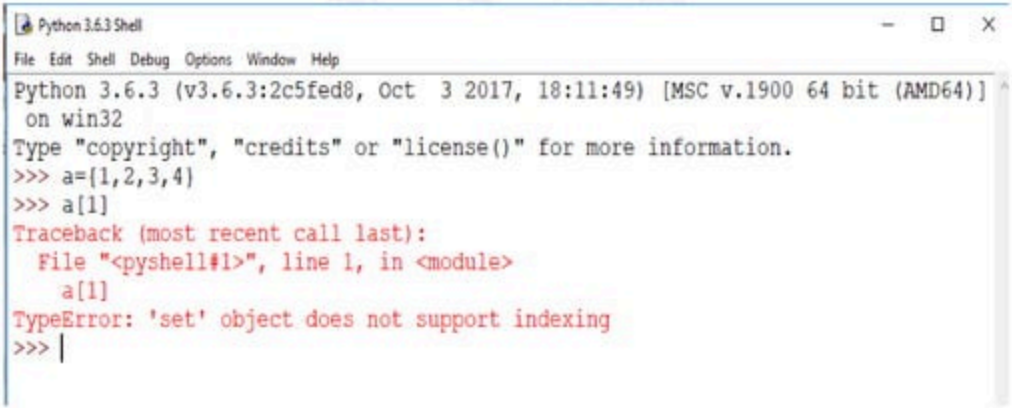
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- Set have unique values. They eliminate duplicates.
- Example:



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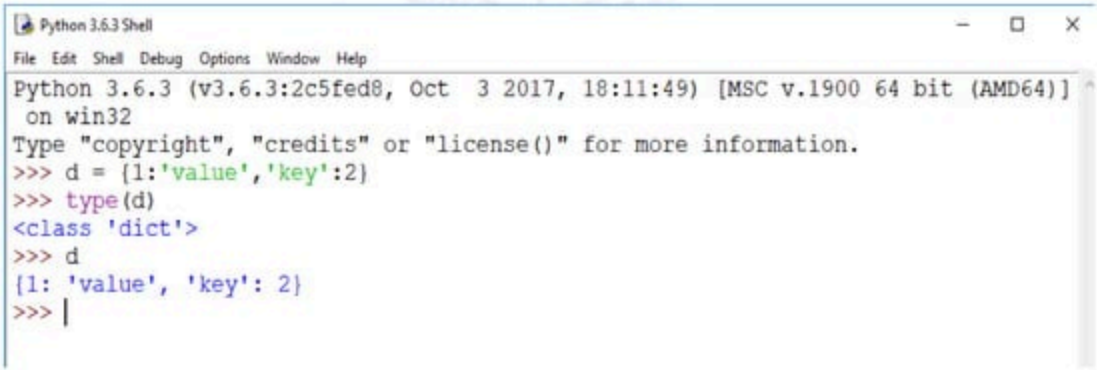
- Since, set are unordered collection, indexing has no meaning.
- Hence the slicing operator [] does not work.
- Example:



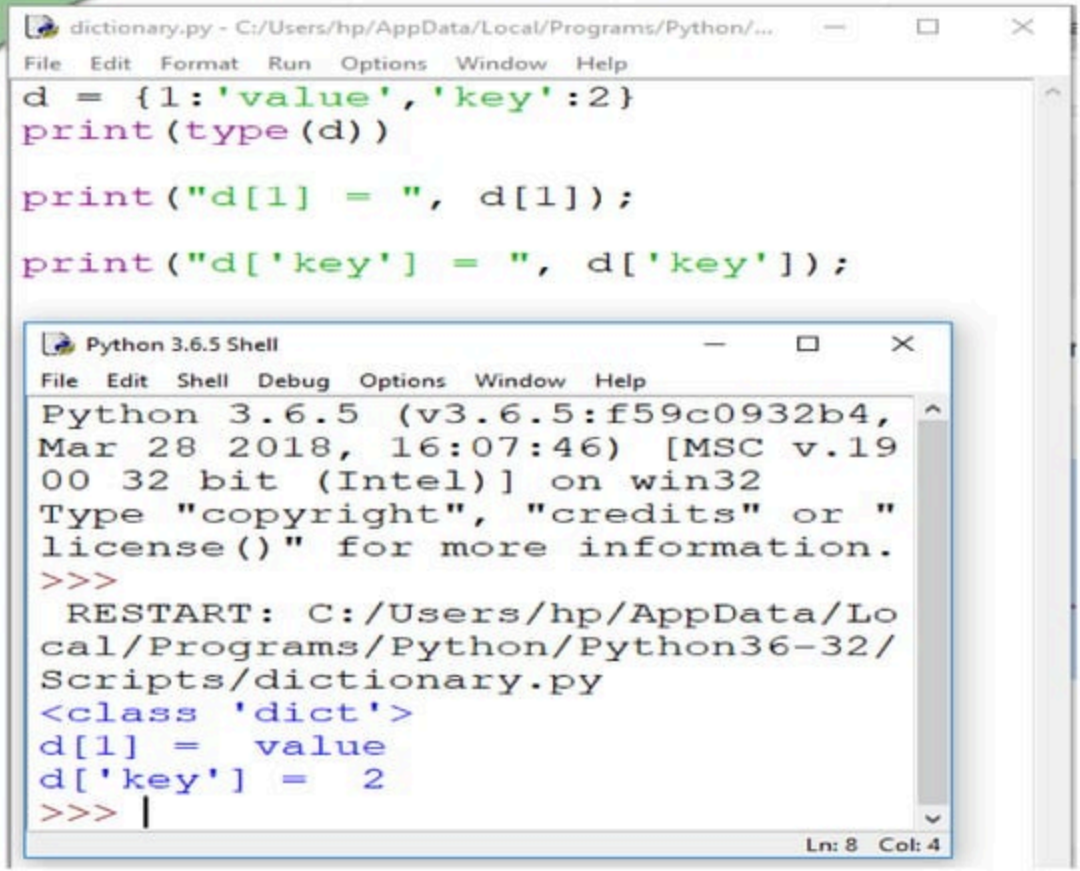
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Python Dictionary

- An unordered collection of key-value pairs.
- Must know the key to retrieve the value.
- Are defined within braces {} with each item being a pair in the form key: value
- Key and value can be of any type



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dictionaryError.py - C:/Users/hp/AppData/Local/P...

File Edit Format Run Options Window Help

```
d = {1:'value','key':2}
print(type(d))

print("d[1] = ", d[1]);

print("d['key'] = ", d['key']);

# Generates error
print("d[2] = ", d[2]);
```

Ln: 10 Col: 0

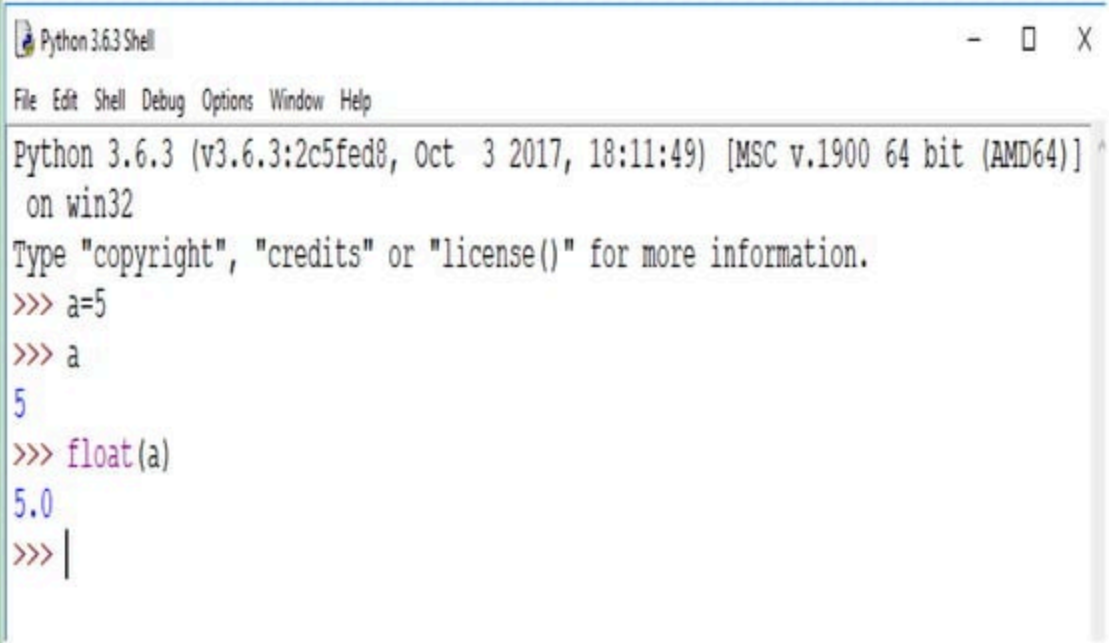
Conversion between data types

- Conversion can be done by using different types of type conversion functions like:
  - int(),
  - float(),
  - str() etc.

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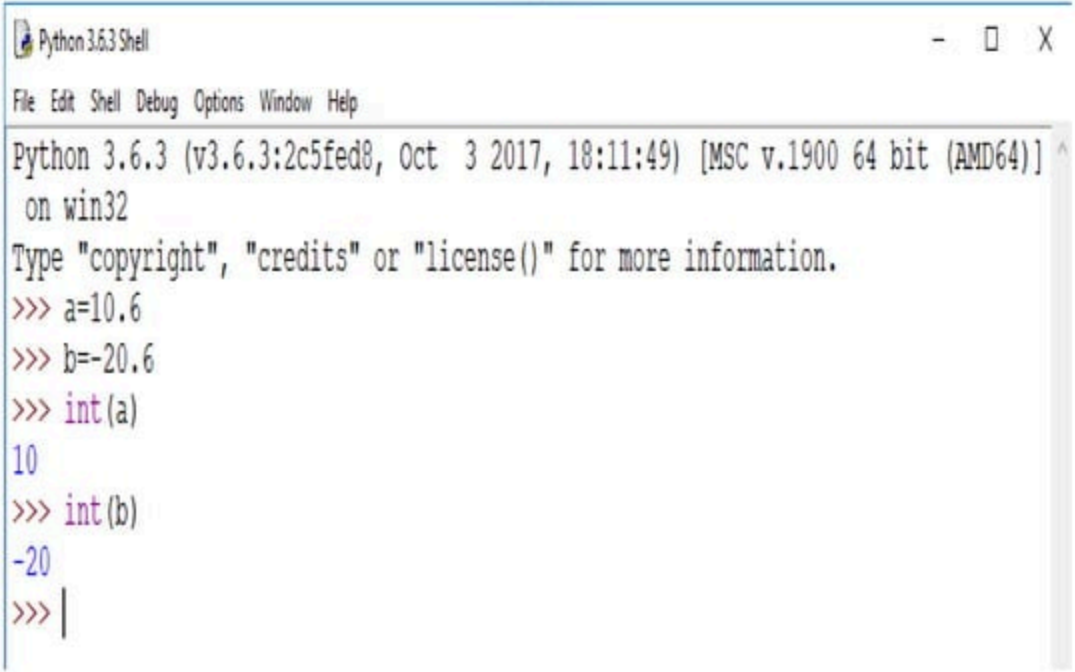
Estd : 1984

int to float



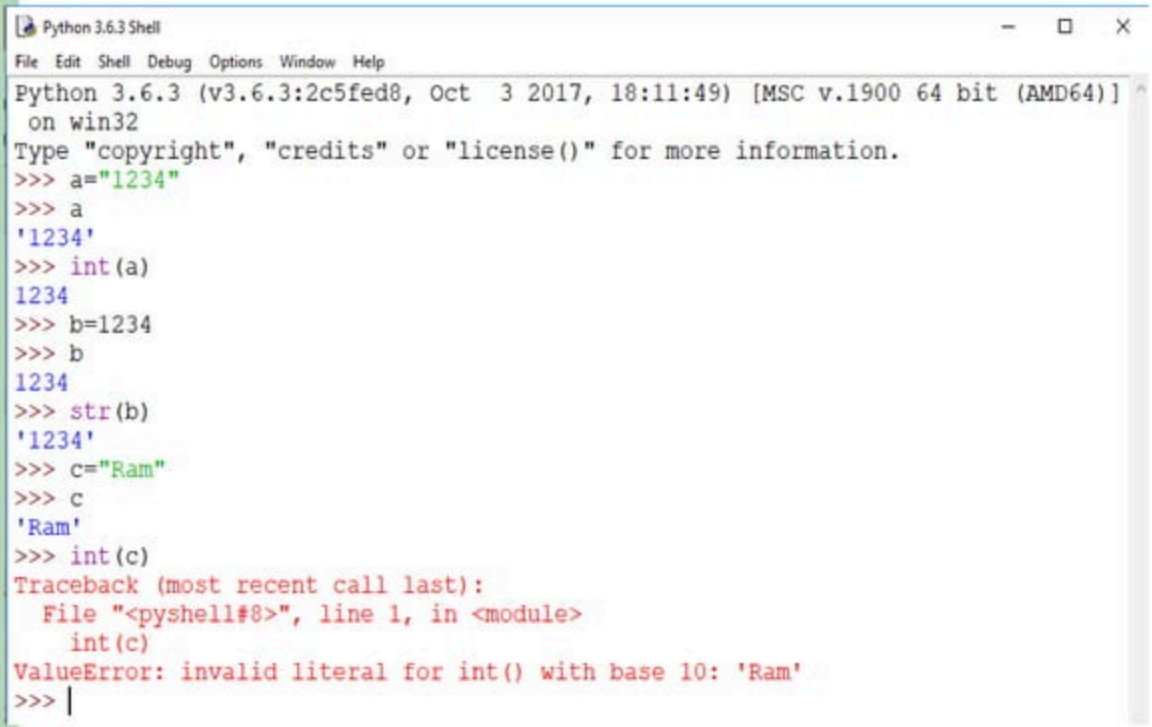
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float to int



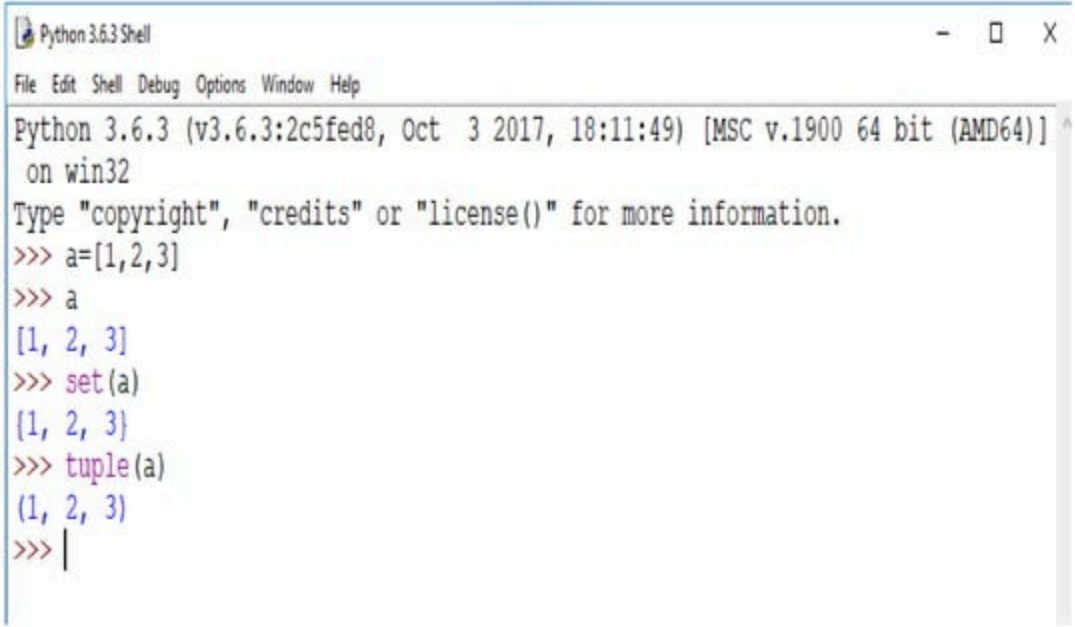
KONGU ENGINEERING COLLEGE

To and form string



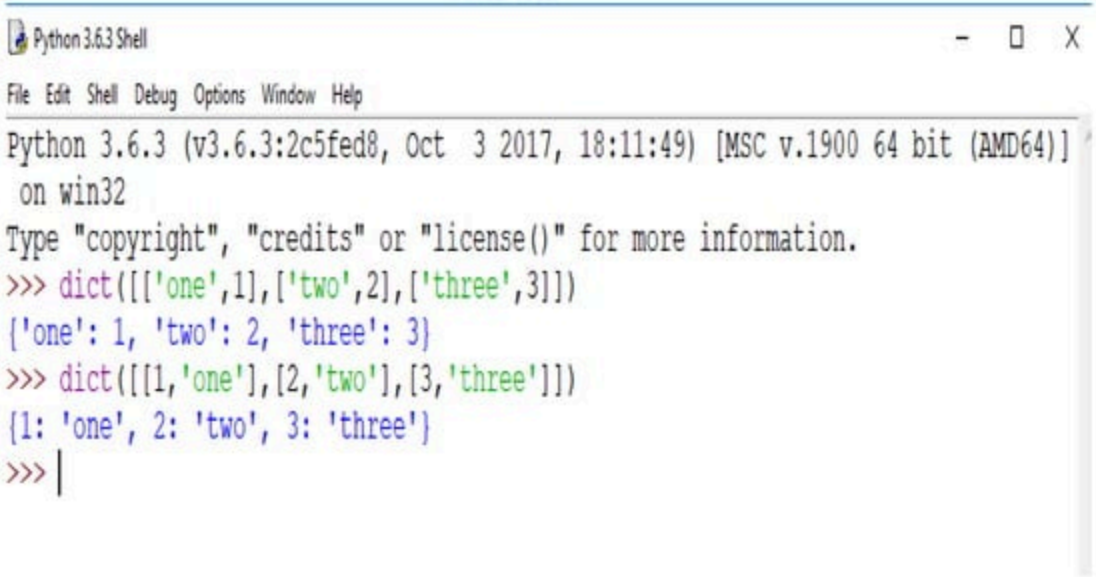
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Convert one sequence to another



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- To convert to dictionary, each element must be a pair:



Python Type Conversion and TypeCasting

- The process of converting the value of one data type (integer, string, float, etc.) to another data type is type conversion
- Python has two types of type conversion.
  - Implicit Type Conversion
  - Explicit Type Conversion

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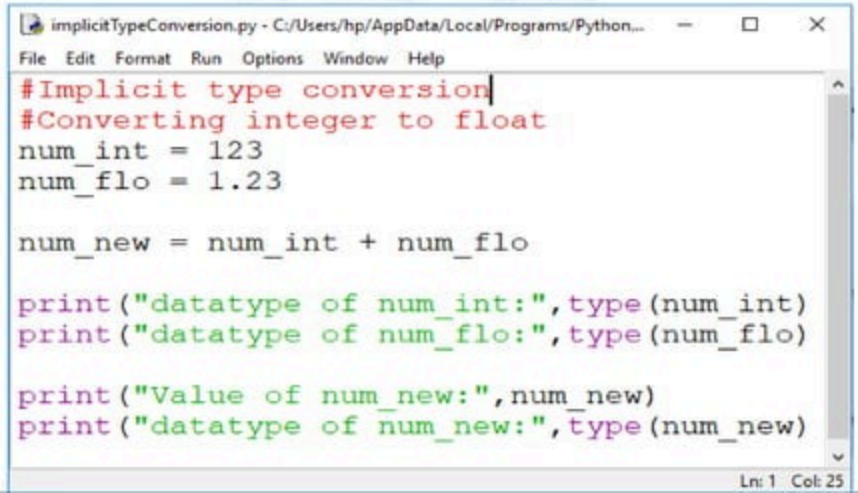
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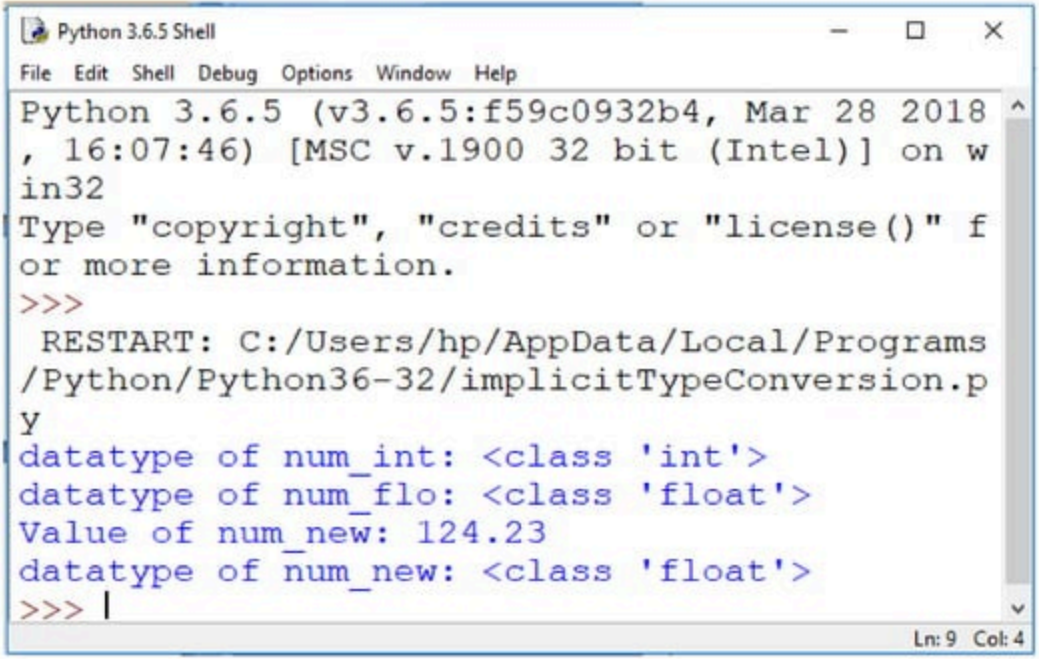
- Type Conversion is the conversion of object from one data type to another data type
- Implicit Type Conversion is automatically performed by the Python interpreter
- Python avoids the loss of data in Implicit Type Conversion
- Explicit Type Conversion is also called Type Casting, the data types of object are converted using predefined function by user
- In Type Casting loss of data may occur as we enforce the object to specific data type

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Implicit Type Conversion

- Automatically converts one data type to another data type
- Always converts smaller data type to larger data type to avoid the loss of data




Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018
, 16:07:46) [MSC v.1900 32 bit (Intel)] on w
in32
Type "copyright", "credits" or "license()" f
or more information.
>>>
RESTART: C:/Users/hp/AppData/Local/Programs
/Python/Python36-32/implicitTypeConversion.p
y
datatype of num_int: <class 'int'>
datatype of num_flo: <class 'float'>
Value of num_new: 124.23
datatype of num_new: <class 'float'>
>>>
Ln: 9 Col: 4

Problem in implicit type conversion

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Untitled

File Edit Format Run Options Window Help

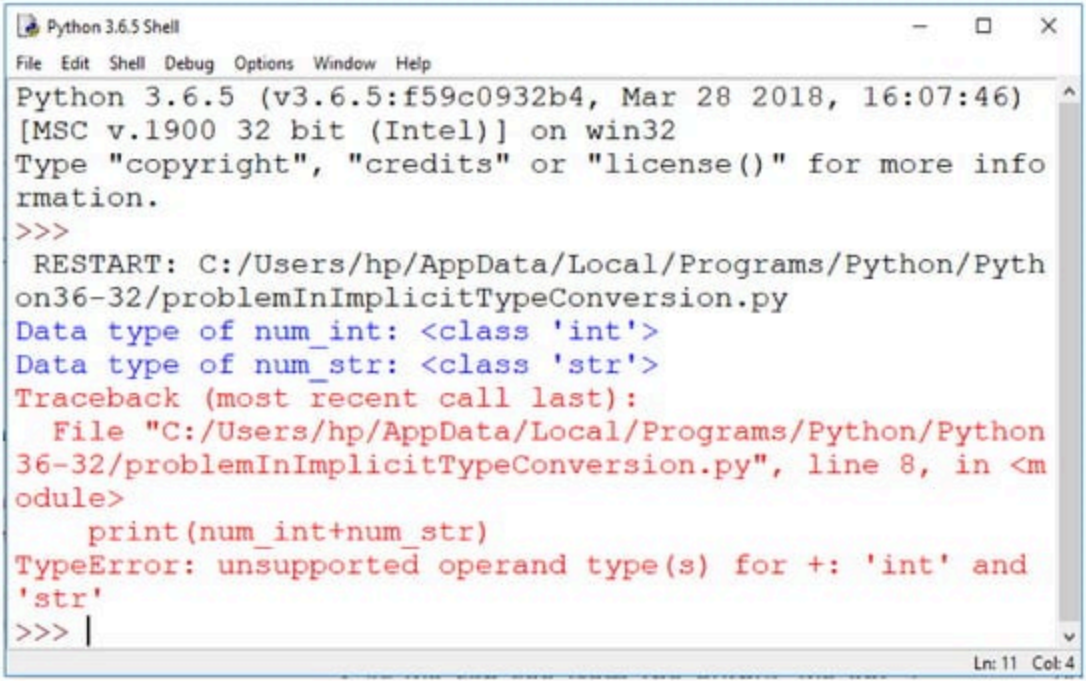
```
#Addition of string(higher) data type and integer(lower) datatype
num_int = 123
num_str = "456"

print("Data type of num_int:",type(num_int))
print("Data type of num_str:",type(num_str))

print(num_int+num_str)

#Python is not able use Implicit Conversion in such condition.
#However Python has the solution for this type of situation
#which is know as Explicit Conversion.
```

Ln: 12 Col: 1


Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46)
[MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more info
rmation.
>>>
RESTART: C:/Users/hp/AppData/Local/Programs/Python/Pyth
on36-32/problemInImplicitTypeConversion.py
Data type of num_int: <class 'int'>
Data type of num_str: <class 'str'>
Traceback (most recent call last):
File "C:/Users/hp/AppData/Local/Programs/Python/Python
36-32/problemInImplicitTypeConversion.py", line 8, in <m
odule>
print(num_int+num_str)
TypeError: unsupported operand type(s) for +: 'int' and
'str'
>>>
Ln: 11 Col: 4

Explicit Type Conversion

- Users convert the data type of an object to required data type.
- Predefined functions like int(), float(), str(), etc. are to perform explicit type conversion
- Is also called typecasting because the user casts (change) the data type of the objects.
- Syntax :
  - (required_datatype)(expression)
  - Int(b)

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Example

explicitTypeConversion.py - C:/Users/hp/AppData/Local/Programs/Python/Python36-32/Scripts/explicitTypeConversion.py (3.6.5)

File Edit Format Run Options Window Help

```
#Addition of string and integer using explicit conversion
num_int = 123
num_str = "456"

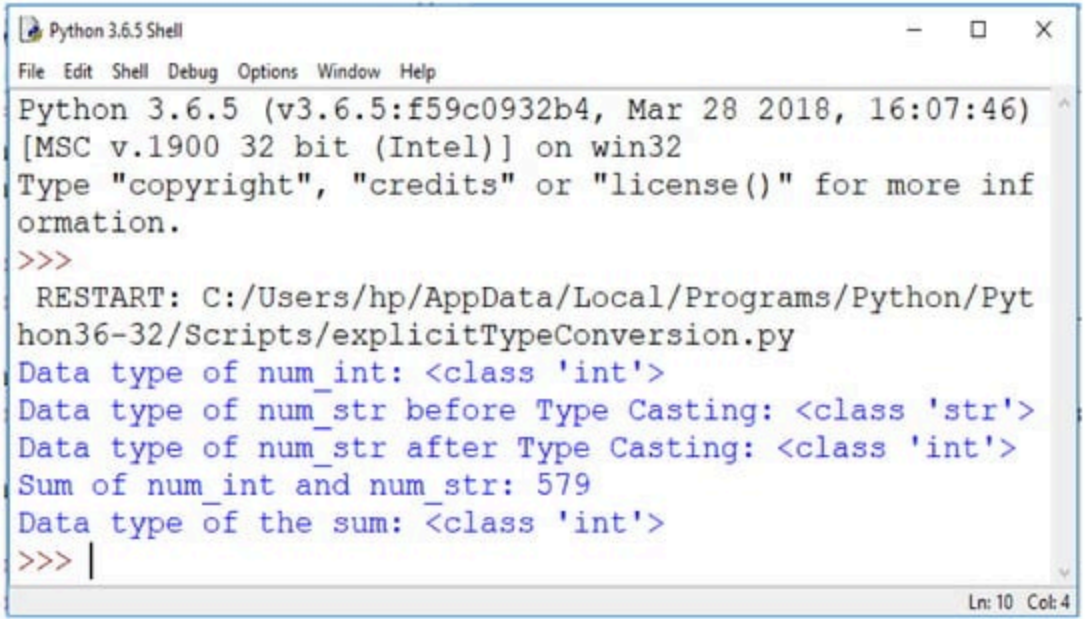
print("Data type of num_int:",type(num_int))
print("Data type of num_str before Type Casting:",type(num_str))

num_str = int(num_str)
print("Data type of num_str after Type Casting:",type(num_str))

num_sum = num_int + num_str

print("Sum of num_int and num_str:",num_sum)
print("Data type of the sum:",type(num_sum))
```

Ln: 1 Col: 57


Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Python 3.6.5 (v3.6.5:f59c0932b4, Mar 28 2018, 16:07:46)
[MSC v.1900 32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more inf
ormation.
>>>
RESTART: C:/Users/hp/AppData/Local/Programs/Python/Pyt
hon36-32/Scripts/explicitTypeConversion.py
Data type of num_int: <class 'int'>
Data type of num_str before Type Casting: <class 'str'>
Data type of num_str after Type Casting: <class 'int'>
Sum of num_int and num_str: 579
Data type of the sum: <class 'int'>
>>>
Ln: 10 Col: 4

Python Input, Output and Import

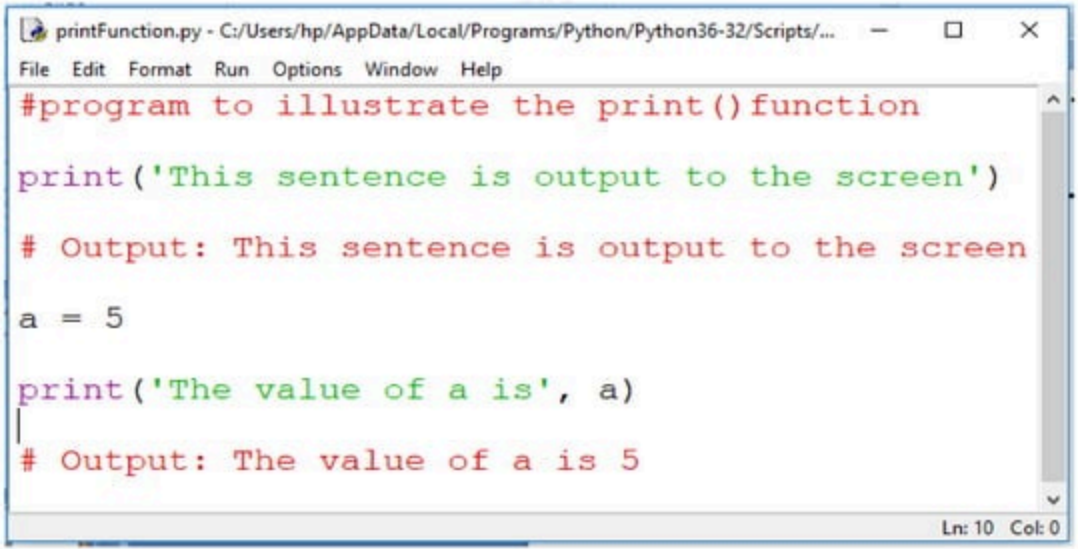
- Widely used functions for standard input and output operations are:
  - print() and
  - input()

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Estd : 1984

Python Output Using print()function

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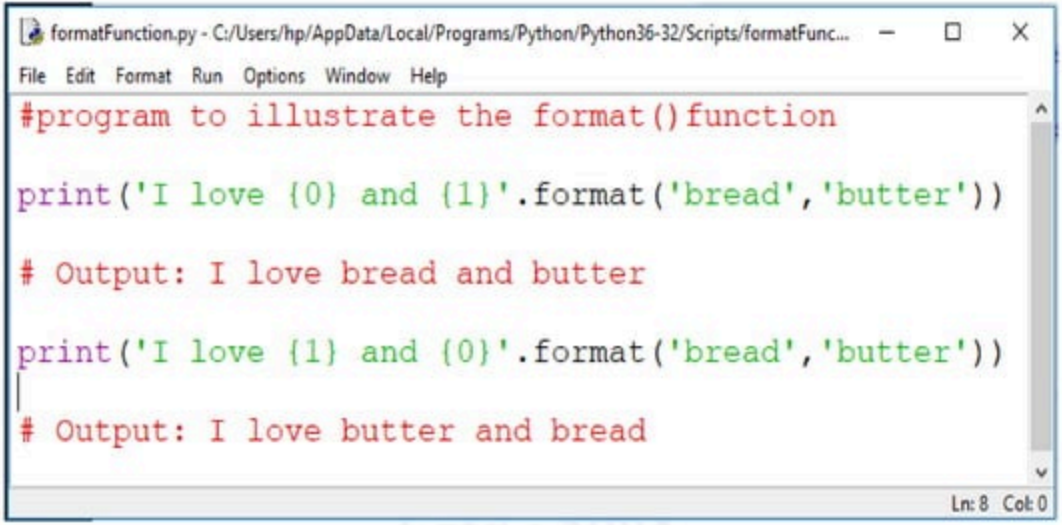
Output formatting

- Can be done by using the str.format() method
  - Is visible to any string object

```
>>> x = 5; y = 10
>>> print('The value of x is {} and y is {}'.format(x,y))
The value of x is 5 and y is 10
>>>
```

- Here the curly braces {} are used as placeholders. We can specify the order in which it is printed by using numbers (tuple index).

Example




```
I love bread and butter
I love butter and bread
>>>
```

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- We can even use keyword arguments to format the string

```
>>> print('Hello {name}, {greeting}'.format(greeting = 'Goodmorning', name = 'Ram'))
Hello Ram, Goodmorning
>>>
```

Estd : 1984

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- We can even format strings like the old printf() style used in C
- We use the % operator to accomplish this

```
>>> x= 12.3456789
>>> print('the value of x is %3.2f' %x)
the value of x is 12.35
>>> print('the value of x is %3.3f' %x)
the value of x is 12.346
>>>
```

Input() function

- input() function to take the input from the user.
- The syntax for input() is :
  - input([prompt])
  - where prompt is the string we wish to display on the screen.
  - It is o

```
>>> num = input('Enter a number: ')
Enter a number: 10
>>> num
'10'
>>>
```

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- Entered value 10 is a string, not a number.
- To convert this into a number we can use int() or float() functions

```
>>> int('10')
10
>>> float('10')
10.0
>>>
```

- This same operation can be performed using the eval(). It can evaluate expressions, provided the input is a string.

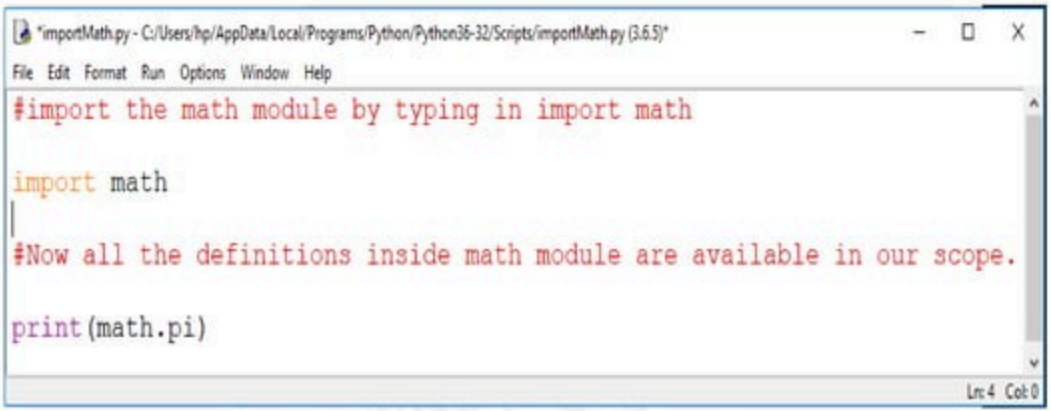
```
>>> int('2+3')
Traceback (most recent call last):
  File "<pyshell#9>", line 1, in <module>
    int('2+3')
ValueError: invalid literal for int() with base 10: '2+3'
>>> eval('2+3')
5
>>>
```

Python Import

- When our program grows bigger, it is a good idea to break it into different modules.
- A module is a file containing Python definitions and statements.
- Python modules have a filename and end with the extension .py
- Definitions inside a module can be imported to another module or the interactive interpreter in Python
- We use the import keyword to do this

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```
3.141592653589793
>>>
```

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- import some specific attributes and functions only, using the from keyword.

```
>>> from math import pi
>>> pi
3.141592653589793
>>>
```

Estd : 1984

Python Operators

- Operators are special symbols in Python that carry out arithmetic or logical computation.
- The value that the operator operates on is called the operand.
- For example:

```
>>> 2+3
5
```

- Here, + is the operator that performs addition.
- 2 and 3 are the operands and 5 is the output of the operation.

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Arithmetic operators

Arithmetic operators in Python

Operator	Meaning	Example
+	Add two operands or unary plus	x + y +2
-	Subtract right operand from the left or unary minus	x - y -2
*	Multiply two operands	x * y
/	Divide left operand by the right one (always results into float)	x / y
%	Modulus - remainder of the division of left operand by the right	x % y (remainder of x/y)
//	Floor division - division that results into whole number adjusted to the left in the number line	x // y
**	Exponent - left operand raised to the power of right	x**y (x to the power y)

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Example

arithmeticOperator.py - C:/Users/hp/AppData/Loc...

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```
# Arithmetic operators in Python
x = 15
y = 4

# Output: x + y = 19
print('x + y =',x+y)

# Output: x - y = 11
print('x - y =',x-y)

# Output: x * y = 60
print('x * y =',x*y)

# Output: x / y = 3.75
print('x / y =',x/y)

# Output: x // y = 3
print('x // y =',x//y)

# Output: x ** y = 50625
print('x ** y =',x**y)
```

Ln: 1 Col: 32

```
x + y = 19
x - y = 11
x * y = 60
x / y = 3.75
x // y = 3
x ** y = 50625
```

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Comparison operators

Comparision operators in Python

Operator	Meaning	Example
>	Greater that - True if left operand is greater than the right	x > y
<	Less that - True if left operand is less than the right	x < y
==	Equal to - True if both operands are equal	x == y
!=	Not equal to - True if operands are not equal	x != y
>=	Greater than or equal to - True if left operand is greater than or equal to the right	x >= y
<=	Less than or equal to - True if left operand is less than or equal to the right	x <= y

comparisionOperator.py - C:/Users/hp/AppData/Local...

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```
#Comparison operators in Python
x = 10
y = 12

# Output: x > y is False
print('x > y  is',x>y)

# Output: x < y is True
print('x < y  is',x<y)

# Output: x == y is False
print('x == y is',x==y)

# Output: x != y is True
print('x != y is',x!=y)

# Output: x >= y is False
print('x >= y is',x>=y)

# Output: x <= y is True
print('x <= y is',x<=y)
```

Ln: 1 Col: 31

```
x > y  is False
x < y  is True
x == y is False
x != y is True
x >= y is False
x <= y is True
```

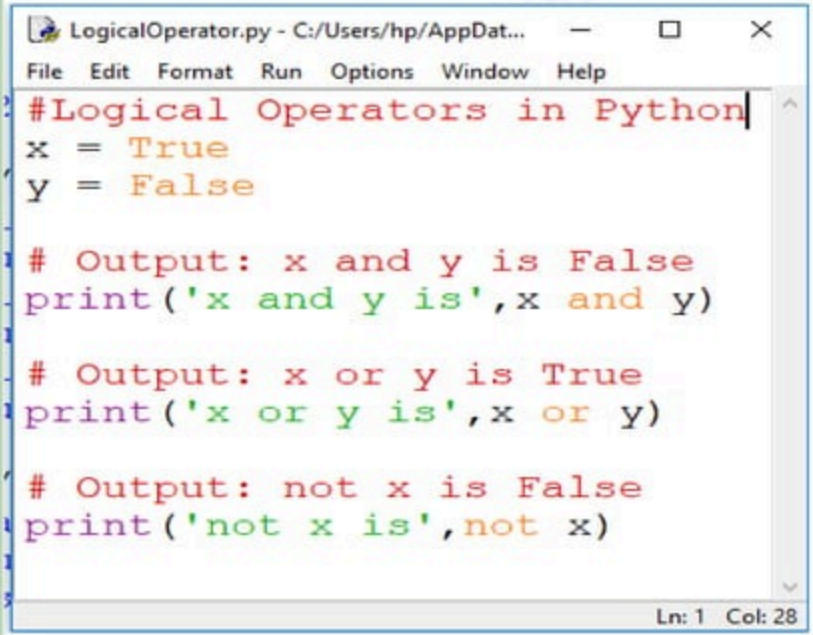
Logical operators

Logical operators in Python

Operator	Meaning	Example
and	True if both the operands are true	x and y
or	True if either of the operands is true	x or y
not	True if operand is false (complements the operand)	not x

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```
x and y is False
x or y is True
not x is False
```

Bitwise operators

- Bitwise operators act on operands as if they were string of binary digits. It operates bit by bit, hence the name.
- For example, 2 is 10 in binary and 7 is 111.
- In the table below: Let x = 10 (0000 1010 in binary) and y = 4 (0000 0100 in binary)

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Bitwise operators in Python

Operator	Meaning	Example
&	Bitwise AND	x& y = 0 (0000 0000)
|	Bitwise OR	x | y = 14 (0000 1110)
~	Bitwise NOT	~x = -11 (1111 0101)
^	Bitwise XOR	x ^ y = 14 (0000 1110)
>>	Bitwise right shift	x>> 2 = 2 (0000 0010)
<<	Bitwise left shift	x<< 2 = 40 (0010 1000)

Assignment operators

- Assignment operators are used in Python to assign values to variables.
- a = 5 is a simple assignment operator that assigns the value 5 on the right to the variable a on the left.
- There are various compound operators in Python like a += 5 that adds to the variable and later assigns the same.
- It is equivalent to a = a + 5.

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Assignment operators in Python

Operator	Example	Equivatent to
=	x = 5	x = 5
+=	x += 5	x = x + 5
-=	x -= 5	x = x - 5
*=	x *= 5	x = x * 5
/=	x /= 5	x = x / 5
%=	x %= 5	x = x % 5
//=	x //= 5	x = x // 5
**=	x **= 5	x = x ** 5
&=	x &= 5	x = x & 5
|=	x |= 5	x = x | 5
^=	x ^= 5	x = x ^ 5
>>=	x >>= 5	x = x >> 5
<<=	x <<= 5	x = x << 5

Special operators

- identity operator
- membership operator

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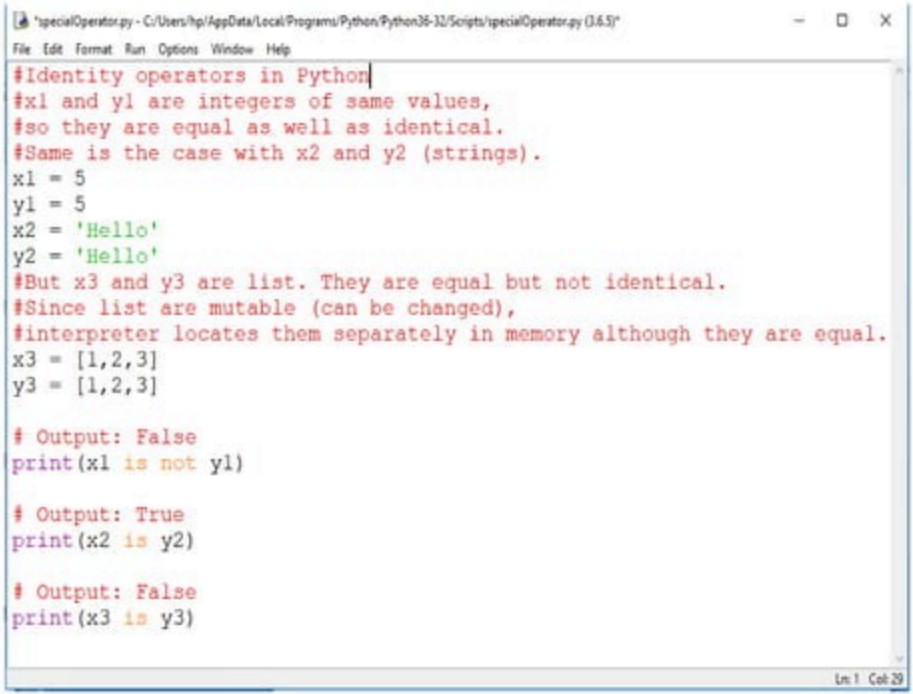
Identity operators

- is and is not are the identity operators in Python.
- They are used to check if two values (or variables) are located on the same part of the memory.
- Two variables that are equal does not imply that they are identical.

Identity operators in Python

Operator	Meaning	Example
is	True if the operands are identical (refer to the same object)	x is True
is not	True if the operands are not identical (do not refer to the same object)	x is not True

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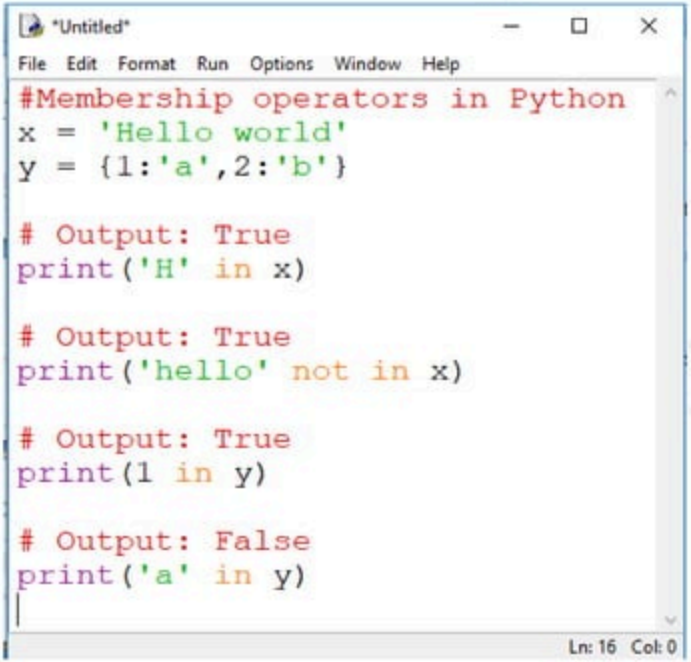

```
False
True
False
```

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Membership operators

- in and not in are the membership operators in Python
- They are used to test whether a value or variable is found in a sequence (string, list, tuple, set and dictionary).
- In a dictionary we can only test for presence of key, not the value

Operator	Meaning	Example
in	True if value/variable is found in the sequence	5 in x
not in	True if value/variable is not found in the sequence	5 not in x




```
True
True
True
False
```

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Thank You!

